

Bipartite Randomized Response Mechanism for Local Differential Privacy

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Abstract

With the increasing importance of data privacy, Local Differential Privacy (LDP) has recently become a strong measure of privacy for protecting each user’s privacy from data analysts without relying on a trusted third party. In this paper, we consider the problem of high-utility differentially private release. Given a domain of items and a distance-defined utility function, our goal is to design a differentially private mechanism that releases an item with the global expected error as small as possible. The recent superior mechanism for this task is the staircase randomized response (SRR) that partitions all candidate items into multiple groups. In this paper, we introduce Bipartite Randomized Response mechanism (BRR), which adaptively divides all candidate items into two parts by utility rankings. In the local search phase, we confirm how many high-utility candidates to be assigned with high release probability, which gives the locally optimal bipartite classification of all candidates. For preserving LDP, the global search phase uniformly selects the smallest number of dynamic high-utility candidates obtained locally. In particular, we give explicit formulas on the uniform number of dynamic high-utility candidates. The global expected error of our BRR can theoretically deliver a decrease with an asymptotically exact ratio. Extensive experiments demonstrate that BRR outperforms the state-of-the-art methods, at up to 23.4% reduction across the standard metrics and datasets. We further show that BRR provides stronger robustness than GRR against baseline poisoning attacks of ranking estimation, yielding more stable rank across diverse data distributions.

1 Introduction

In recent years, as concerns over data privacy have intensified [27], Differential Privacy (DP) has emerged as a vital approach for protecting individual information [5]. DP, first introduced by Dwork in 2006 [8], addresses the issue of privacy leakage caused by minor changes in the data source. With rigorous theoretical guarantees, centralized DP ensures

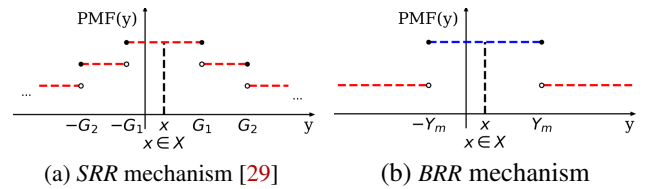


Figure 1: Probability mass function (PMF) for SRR and BRR.

that the output information remains influenced by any single record only within a specified threshold, thus preventing third parties from inferring sensitive information based on output variations. Users have to provide original data to a central server, which acts as a trusted entity to preserve centralized DP. In contrast, Local Differential Privacy (LDP) [16] allows users to send data with added noise directly to the central server. This ensures that individual privacy is protected before the data upload.

One foundational mechanism in LDP is the Randomized Response (RR) mechanism introduced by Warner [34]. While RR is effective for protecting binary data, its applicability is limited to scenarios where the answer domain consists of only two values. To address this, the Generalized Randomized Response mechanism (GRR, also known as k -RR) [15] is proposed. GRR extends the RR mechanism to accommodate larger answer domains consisting of N possible candidates. Under GRR, the data publisher selects and reports the true value with a fixed probability p , while assigning an equal probability to each of the remaining $N - 1$ values with probability $\frac{1-p}{N-1}$. This extension retains the privacy-preserving properties of RR while broadening its applicability to use cases involving multi-valued categorical data, such as user preferences. It is primarily designed and analyzed as a primitive for frequency estimation [10, 31], and can also be used directly as a competitive mechanism for a range of tasks, including data collection [37], decision tree [20], and deep learning [21]. The GRR mechanism is order optimal in the low privacy regime (large ϵ) for a series of utility functions [15]. Particularly,

some LDP mechanisms have been deployed in the location-based service (LBS) applications (e.g., Google, Baidu and Microsoft) but not effective on performance, e.g., L-SRR realizing Staircase Randomized Response (SRR) [29] in LBSs as shown in Fig. 1(a). There exist some limitations on private data release mechanisms in the LBS scenarios as follows:

- Candidates usually have different degrees of utility. SRR partitions all locations into multiple groups for staircase probabilities as in Fig. 1(a), often giving poor utility.
- Candidates within the answer domain are often numerous. With the increasing number of candidates, the quality loss of GRR grows explosively as shown in Fig. 5(a).
- Users sometimes require high privacy levels (small ϵ) for protecting sensitive data. However, GRR can not achieve optimal utility in this case [15] as well as SRR.

To address these limitations, one has to adjust the probability distribution on candidates for achieving higher utility. This raises the open question of how to make the adjustment efficient with theoretical guarantees. In this paper, we propose the *Bipartite Randomized Response* (BRR) mechanism for the task of differentially private release. BRR incorporates the utility of candidates in the answer domain and consists of the local and global search phases. Starting with the GRR framework, BRR adjusts the probabilities assigned to each candidate based on their utility to the true answer.

Let X denote the space of true values and Y the space of released values, where Y is identical to $X = \{1, 2, \dots, N\}$. When perturbing a specific x_i , BRR assigns higher publishing probabilities to the m candidates most close to x_i (with higher utility) as shown in Fig. 1(b). By adjusting the distribution of publishing probabilities, BRR increases the likelihood of publishing data that are closer to the true item while decreasing the likelihood for data that are farther. This reallocation mechanism renders the publishing probabilities more rational.

We can theoretically demonstrate that BRR significantly reduces utility loss, cf. Theorem 3 in the equidistant point domain. Fig. 5(a) shows that BRR has a significantly lower query loss ($QLoss$) compared to GRR with varying N . In addition, when N tends to infinity, the asymptotically exact ratio for quality-loss (and global expected-error) of BRR over N is verified in Fig. 5(b).

BRR enjoys the same desirable properties of the GRR, and its local and global expected errors are never higher in any privacy region, but can be exponential times (dependent on privacy budget ϵ) lower as long as the count N of candidates is sufficiently large. Furthermore, we show that in reasonable settings no better mechanism exists: the BRR mechanism is optimal in a reasonable “global” sense, independent of the prior distribution as compared to L-SRR and GRR in Fig. 9.

The **main contributions** of this work are summarized as follows:

- We propose the *Bipartite Randomized Response* (BRR) mechanism as an alternative to SRR and GRR mechanisms for the task of data release while aiming to obtain global high utility under strict LDP constraints.
- BRR mechanism is conducted through a two-phase approach. In the first phase, given priori item, we sort the candidates according to their utilities and divided into two parts for maximizing local utility. In the second phase, for enforcing LDP the global search selects uniformly the smallest count among dynamic high-utility parts obtained locally.
- For the BRR mechanism in the equidistant point domain, we present explicit formulas for the global uniform number of elements that make up the dynamic high-utility part. The expected error of BRR offers a decrease with an asymptotically exact ratio as elaborated later in Theorem 3.
- A series of experiments are executed for comparisons and applications on public datasets. The significant utility improvements over the state-of-the-art methods in both discrete and continuous domains have impacts on practical deployments of local differential privacy and the privacy-utility trade-off in social service platforms.

Roadmap. In Section 2, some preliminaries are presented. Section 3 expounds our BRR mechanism in the equidistant point domain. Section 4 discusses the exploration of BRR in general domains. Section 5 presents some comparative experiments on BRR and the state-of-the-art response mechanisms, and evaluates the robustness against poisoning attacks. Section 6 reviews the related work. Finally, we offer a comprehensive conclusion of this paper. Some detailed proofs and evaluations are given in Appendix.

2 Preliminaries

In this section, we delve into the concept of local differential privacy (LDP) and associated noise-adding mechanisms, including generalized randomized response (GRR), exponential mechanism, Laplace mechanism, staircase randomized response (SRR), and geometric mechanism. These methods constitute pivotal techniques for implementing differential privacy, and each will be elucidated in detail below. Table 1 summarizes the main notations in this paper.

2.1 Local Differential Privacy

Classical (centralized) DP and Local DP (LDP) are both privacy-preserving mechanisms used to protect sensitive data. While both methods aim to ensure privacy, they differ in their implementation and application scenarios [17]. Let $D \sim D'$ denote the neighboring relation between D and D' , that is,

Table 1: Summary of Notations

Symbol	Definition
$\mathcal{X}$	the domain of priori values or points
$\mathcal{Y}$	the domain of reported values (points), identical to $\mathcal{X}$
N	the cardinality of domain $\mathcal{Y}$, i.e., $N = \mathcal{Y} $
Y_m	the set of m high probability e^ϵ -weighted items
ϵ	privacy budget
p	probability of publishing the true item in GRR
m	the uniform number of high-weighted items
p_m^*	prob. of reporting each top- m utility item in BRR
q_m^*	probability of reporting each low utility item in BRR
k	label of the true item, i.e., the k -th priori item
λ_i	the i -th highest utility of items given the true item
$\lambda_i^{(k)}$	the i -th highest utility of items for the k -th priori item
s_i	weight of reporting the λ_i -utility item, with $s_1 = e^\epsilon$
$s_i^{(k)}$	weight of reporting the item with $\lambda_i^{(k)}$ -utility
w_i	normalized probability weight s_i , with $\sum_i w_i = 1$
$w_i^{(k)}$	normalized probability weight of $s_i^{(k)}$
Q	local utility function (expected error) by $Q = \sum_i \lambda_i w_i$
π	the prior probability distribution on the domain
$Q^{(k)}$	local expected error for the k -th priori item, Eq. (13)
Q_g	global utility function (expected error), Eq. (14)
$QLoss$	global service quality loss defined by Eq. (16)

the datasets D and D' differ on adding or deleting only one element for unbounded DP [17].

Centralized ϵ -DP provides strong privacy guarantees on datasets under the assumption of a trusted data collector. However, in many practical scenarios, such as mobile or edge computing environments, users may not be willing to trust a central server holding their raw data. To address this concern, a stronger notion, Local Differential Privacy (LDP), is formulated in [16]. We will use the definition of LDP given by Erlingsson et al. [9].

Definition 2.1 (ϵ -Local Differential Privacy). *A randomized mechanism $\mathcal{M}$ satisfies ϵ -LDP, where $\epsilon \geq 0$, if and only if for any input $x, x' \in \mathcal{X}$, we have*

$$\forall y \in \mathcal{M}(\mathcal{X}) : \Pr[\mathcal{M}(x) = y] \leq e^\epsilon \cdot \Pr[\mathcal{M}(x') = y]. \quad (1)$$

Here, $\mathcal{M}(\mathcal{X})$ denotes the set of all possible outputs of $\mathcal{M}$.

A key quantity in designing mechanisms that satisfy either ϵ -DP or ϵ -LDP is the sensitivity of a scoring function [8]. It captures the maximum influence that a single individual can have on the result of a numeric query.

Definition 2.2 (Sensitivity). *Let $D \sim D'$. The sensitivity of a scoring function q , denoted by Δq , is given by*

$$\Delta q = \max_{D \sim D'} |q(D) - q(D')|. \quad (2)$$

2.2 Laplace Mechanism

The Laplace Mechanism [7, 8] is the most widely used obfuscation mechanism for DP, which accomplishes privacy protection by adding noise drawn from a Laplace distribution to the query results.

Definition 2.3 (Laplace Mechanism). *For a given query function q , the noisy output is defined as:*

$$\mathcal{M}(D) = q(D) + \text{Lap}\left(\frac{\Delta q}{\epsilon}\right). \quad (3)$$

Here, $\text{Lap}(\frac{\Delta q}{\epsilon})$ denotes a Laplace distribution with a mean of 0 and scale parameter $\frac{\Delta q}{\epsilon}$.

2.3 Exponential Mechanism

The Exponential Mechanism, introduced by McSherry and Talwar [22], is a versatile DP mechanism suitable for arbitrary query ranges. The fundamental principle involves selecting outputs from the candidates space based on a score function.

Definition 2.4 (Exponential Mechanism). *For any quality function $q : \mathbb{D} \times \mathcal{O} \rightarrow \mathbb{R}$, $o \in \mathcal{O}$, the Exponential Mechanism $\mathcal{M}(D)$ outputs o with probability,*

$$\Pr[\mathcal{M}(D) = o] \propto \exp\left(\frac{\epsilon q(D, o)}{2\Delta q}\right). \quad (4)$$

2.4 Generalized Randomized Response (GRR)

Local privacy dates back to Warner [34], who introduced the Randomized Response (RR, also known as W-RR) mechanism to deal with binary responses. Generalized Randomized Response (GRR, also known as k -RR) is a generalization of the RR to handle multiple choice problems by Kairouz et al. [15].

Let $\mathcal{X}$ denote the input domain consisting of $N \geq 2$ elements. GRR maps $\mathcal{X}$ stochastically onto itself (i.e., $\mathcal{Y} = \mathcal{X}$). For any $x \in \mathcal{X}$ and $y \in \mathcal{Y}$, the perturbation probability distribution is defined by

$$\Pr[y|x] = \begin{cases} \frac{e^\epsilon}{e^\epsilon + N - 1} & \text{if } y = x, \\ \frac{1}{e^\epsilon + N - 1} & \text{if } y \neq x. \end{cases} \quad (5)$$

If $N = 2$, this gives the definition of RR mechanism.

2.5 Staircase Randomized Response (SRR)

Given the domain $\mathcal{X}$, for any input $x \in \mathcal{X}$, all the possible output locations can be partitioned into m groups $G_1(x), \dots, G_m(x)$ based on their distances to x .¹ Note that, the

¹W.l.o.g., the distances from x to locations in $G_j(x)$ are farther if j is larger. The closest group is $G_1(x)$ whereas the farthest group is $G_m(x)$.

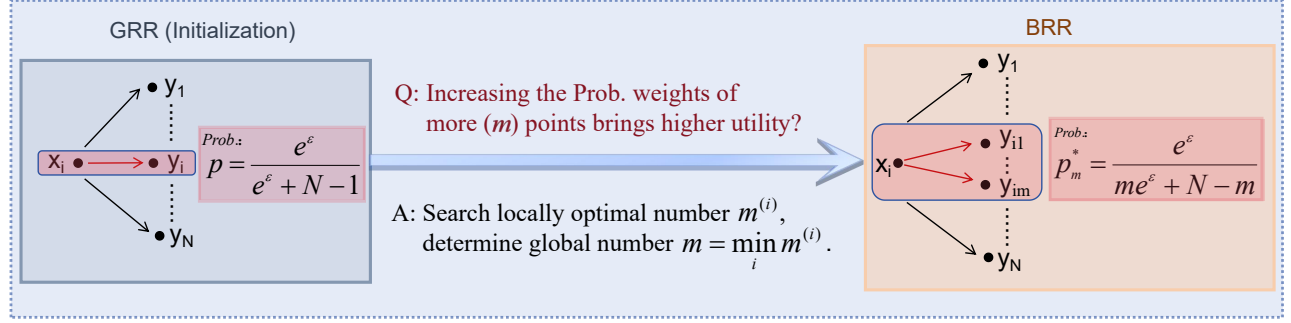


Figure 2: Framework of BRR mechanism. Beginning with GRR, we maximize the pointwise prob. weight (up to e^ϵ) of high-quality points by Alg. 1 for obtaining locally optimized $k = m^{(i)}$ of each priori point x_i . Then the minimal $m^{(i)}$ presents the globally uniform m of e^ϵ -weighted candidates (with prob. p_m^*), while the others are 1-weighted, for the final BRR, cf. Alg. 2.

partitioning $G_j(x)$ is dependent on the input location x . For each input location x , the Staircase Randomized Response mechanism [29] defines the perturbation probabilities (for perturbing x to any output location y) for each location group $G_i(x)$ by

$$q(y|x) = \begin{cases} \alpha_1(x), & \text{if } y \in G_1(x) \\ \vdots & \vdots \\ \alpha_m(x), & \text{if } y \in G_m(x) \end{cases} \quad (6)$$

to satisfy ϵ -differential private, where $\alpha_1(x), \dots, \alpha_m(x)$ are the distance-based perturbation probabilities for locations in m different groups perturbed from $\mathcal{X}$, and the gap between the perturbation probabilities in every adjacent groups is the same in $\alpha_1(x), \dots, \alpha_m(x)$, as shown in Fig. 1(a).

2.6 Geometric mechanisms

The (linear) geometric mechanism [11] maps the space of integers $\mathbb{Z}$ to itself with stochastic probability. To be specific, given $\epsilon > 0$, every $y \in \mathbb{Z}$ is mapped to $z \in \mathbb{Z}$ with probability

$$P(z|y) = ce^{-\epsilon|z-y|} \quad \text{where } c = \frac{1 - e^{-\epsilon}}{1 + e^{-\epsilon}}. \quad (7)$$

The well-known *truncated* version of the geometric mechanism is defined on a bounded range of integers. Let r_1, r_2 be two integers with $r_1 < r_2$. Then the mechanism maps every y between r_1 and r_2 into an integer z in the same range with probability

$$P(z|y) = c_z e^{-\epsilon|z-y|}, \quad (8)$$

where

$$c_z = \frac{1}{1 + e^{-\epsilon}} \begin{cases} 1 & \text{if } z \in \{r_1, r_2\}, \\ 1 - e^{-\epsilon} & \text{if } r_1 < z < r_2, \\ 0 & \text{otherwise.} \end{cases} \quad (9)$$

Then the geometric mechanism satisfies $(r_2 - r_1)\epsilon$ -LDP.

2.7 Design Consideration

We consider a scenario where the user wants to protect the privacy of her/his actual value x_i from a specified discrete domain $\mathcal{X} = \{1, 2, \dots, N\}$, by reporting a pseudo-value y_i in a finite set $\mathcal{Y}$, where usually $\mathcal{Y} = \mathcal{X}$. It is desirable to develop an obfuscation mechanism that satisfies ϵ -differential privacy and generates perturbed values with effective performance.

Naturally, beginning with GRR by (5), we first search for locally optimized high-quality points whose number $m^{(i)}$ is due to the starting priori point x_i . Then we confirm the uniform number m of high probability e^ϵ -weighted candidates for the desired BRR mechanism.

On the construction of BRR, we have to overcome five challenges as follows:

- In the local search phase, why are the optimal distributed probability weights simply bipartite, only 1 and e^ϵ ?
- In the global search phase, how shall we confirm the uniform number m of high probability e^ϵ -weighted candidates that is independent of the priori distribution?
- Can we compute the uniform number m by some explicit formula?
- What is the impact of our proposed BRR mechanism on theory and applications? Specifically, to what extent does the BRR outperform SOTA methods in release utility?
- Can BRR maintain robustness against poisoning attacks in ranking estimation while improving data utility?

3 Design of BRR in equidistant point domain

In this section, we consider the one-dimensional domain that consists of N equidistant points on the real line. Without loss of generality, we assume $\mathcal{X} = \{1, 2, \dots, N\}$ and denote the data utility by the expected error/distance between the true

Algorithm 1 Local highest-utility response search algorithm

Input: quality loss $0 \leq \lambda_1 \leq \lambda_2 \leq \dots \leq \lambda_N$, privacy budget ϵ

- 1: Initialize: $s_1 = e^\epsilon, s_2 = \dots = s_N = 1, w_1 = \frac{e^\epsilon}{e^\epsilon + N - 1}, w_2 = \dots = w_N = \frac{1}{e^\epsilon + N - 1}, Q = \sum_i \lambda_i w_i, m = 1$
- 2: **for** $i = 2$ **to** N **do**
- 3: Compute $\frac{\partial Q}{\partial s_i} = \frac{\sum_j (\lambda_i - \lambda_j) s_j}{(\sum_j s_j)^2}$
- 4: **if** $\frac{\partial Q}{\partial s_i} < 0$ **then** $s_i = e^\epsilon$ and $m = i$, **otherwise**, break
- 5: **end for**
- 6: $w_i = \frac{s_i}{\sum_j s_j}$ **for** $i = 1$ **to** N

Output: $(w_1, \dots, w_N)$ and m

and reporting/pseudo points. The longer distance gives the lower data utility.

In the following, we first introduce the basic expression of BRR as illustrated by Fig. 2. Then we focus on determining the number m of high weighted candidates in the BRR mechanism. This includes the local and global search phases for m .

3.1 Bipartite Randomized Response (BRR)

Given a priori point x , privacy budget ϵ and the uniform number m of high probability e^ϵ -weighted candidates, the set of m points is denoted by Y_m . Clearly, $Y_m \subset Y$ depends heavily on the priori point x .

As shown in Fig. 2, we introduce Bipartite Randomized Response (BRR) mechanism whose perturbation probability distribution is defined by

$$\Pr[y|x] = \begin{cases} \frac{e^\epsilon}{me^\epsilon + N - m} & \text{if } y \in Y_m, \\ \frac{1}{me^\epsilon + N - m} & \text{if } y \notin Y_m. \end{cases} \quad (10)$$

When $m = 1$, BRR becomes the initial trivial mechanism GRR. In addition, following such a definition, the BRR satisfies differential privacy. Under this restriction, we will investigate how to search the optimal number m for making the utility of the BRR mechanism as high as possible. This approach includes two phases: the local and global search phases.

3.2 Determining Locally Optimal m for BRR

For local publishing, given the actual point x from the domain $\mathcal{X}$, let λ_i denote the data utility of reporting each point x_i in $\mathcal{X}$. The utility can be defined adaptively in various scenarios such as the Jaccard similarity discussed in Section B.1. Currently, we assign simply the Euclidean distance as the metric of data utility, which can also be called service quality loss. Then, the closer reporting point to the actual point x gives higher utility and smaller quality loss. After sorting, the quality loss of all

points are increasingly like

$$0 = \lambda_1 \leq \lambda_2 \leq \dots \leq \lambda_N,$$

with the pairs $(y_1, \lambda_1), \dots, (y_N, \lambda_N)$. Accordingly, the initial probability distribution weights s_i can satisfy

$$e^\epsilon = s_1 \geq s_2 \geq \dots \geq s_N = 1.$$

By normalization of the sum, define $w_i = s_i / \sum_j s_j$ such that $\sum_i w_i = 1$. The local expected error of reporting is defined by $Q = \sum_i \lambda_i w_i$, i.e.,

$$Q = \frac{\sum_i \lambda_i s_i}{\sum_j s_j}. \quad (11)$$

To further explore the effect of s_i on Q , we take the partial derivative of Q with respect to s_i as

$$\frac{\partial Q}{\partial s_i} = \frac{\lambda_i \sum_j s_j - \sum_j \lambda_j s_j}{(\sum_j s_j)^2} = \frac{\sum_j (\lambda_i - \lambda_j) s_j}{(\sum_j s_j)^2}. \quad (12)$$

It can be observed that the monotonicity of Q with respect to s_i is independent of the value of s_i . As illustrated in Algorithm 1, we perform the local search to minimize the expected error Q .

Step 1: Given the actual point x , initialize the search algorithm as the GRR mechanism, i.e., $s_1 = e^\epsilon$, and the other, $s_2, \dots, s_N$, are uniformly assigned the minimum weight value 1.

Step 2: Increase s_2 to the maximum value $s_1 = e^\epsilon$, then successively increase $s_3, s_4, \dots$, until some s_i satisfies $\frac{\partial Q}{\partial s_i} \geq 0$. Stop iterating at this point y_i , fix $m = i$ and $Y_m = \{y_1, \dots, y_m\}$ and remain $s_i, \dots, s_N$ unchanged to be 1.

Step 3: Update weights w_j by (10) and finish local search phase.

Following this, different priori points could give different locally optimal m and further different perturbation probability distributions by employing (10). This will certainly violate ϵ -differential privacy guarantee. Then, we have to search for a uniform number m , even independent of the prior distribution over the domain $\mathcal{X}$ in the next section.

3.3 Determining Globally Uniform m for BRR

For global publishing with differential privacy guarantee, given each real occurrence $k \in \mathcal{X}$, i.e., $1 \leq k \leq N$, we first sort the quality losses $\lambda_i^{(k)}$ and initially distributed probability weights $s_i^{(k)}$ of all points as follows,

$$0 = \lambda_1^{(k)} \leq \lambda_2^{(k)} \leq \dots \leq \lambda_N^{(k)}, \quad e^\epsilon = s_1^{(k)} \geq s_2^{(k)} \geq \dots \geq s_N^{(k)} = 1.$$

The weights $s_i^{(k)}$ are normalized to generate $w_i^{(k)} = \frac{s_i^{(k)}}{\sum_j s_j^{(k)}}$ with satisfying $\sum_i w_i^{(k)} = 1$. Note that for different occurrence k , the sequence of pairs $\{(y_1, \lambda_1^{(k)}), \dots, (y_N, \lambda_N^{(k)})\}$ varies.

For the occurrence item k , the initial weight vector $s^{(k)}$ is the same as the GRR, where the first weight $s_1^{(k)} = e^\epsilon$ (for the point k itself) and the remaining weights are 1. In particular, the first pair is $(\lambda_1^{(k)}, s_1^{(k)}) = (0, e^\epsilon)$.

The local expected error (local quality loss) for the occurrence item k is denoted by

$$Q^{(k)} = \sum_i \lambda_i^{(k)} w_i^{(k)}. \quad (13)$$

By averaging uniformly the local utility of all points, the global expected error (global utility loss function) is denoted by, with uniform average,

$$Q_g = \frac{1}{N} \sum_k Q^{(k)} = \frac{1}{N} \sum_k \sum_i \lambda_i^{(k)} w_i^{(k)}. \quad (14)$$

As illustrated in Algorithm 2, we perform the global search to minimize the quality loss Q_g . This gives the sequence $(w_1, \dots, w_N)$ uniformly for any priori k , then $w_i^{(k)}$ can be simply rewritten as w_i in (13) and (14). The search steps are summarized as follow.

Step 1: Search locally optimal $m^{(k)}$ for each occurrence k by Algorithm 1, cf. Lines 1-5.

Step 2: Search globally uniform $m = \min_k m^{(k)}$, cf. Line 6.

Step 3: Update all weights w_j by (10) where Y_m is replaced by $Y_m^{(k)}$ that contains the first m points closest to the point (occurrence) k , and finish the global search phase, cf. Line 7.

Starting from the second weight $s_2^{(k)}$, each weight is adjusted sequentially. During the adjustment process, the partial derivative of utility function $Q^{(k)}$ with respect to $s_i^{(k)}$ is calculated to determine whether increasing the current weight can improve the utility.

Once the weights of all occurrence items k have been optimized, we compare the optimal $m^{(k)}$ derived from each local search phase and select the minimum value as the global final threshold. This allows us to construct a unified global BRR mechanism. In this global BRR mechanism:

- The number m of high weighted candidates is chosen as the smallest $m^{(k)}$ across all occurrences. This ensures that the obfuscation mechanism achieves higher utility than GRR at each occurrence point k . If m increases, the local utility of publishing some occurrences would certainly decrease. In addition, this selection is independent of the prior distribution π over the domain $\mathcal{X}$.
- For each local occurrence k , the first m points closest to the actual occurrence point are assigned a probability weight of e^ϵ , while the remaining points with weight of 1 before normalization. Thus, the so-called set Y_m depends on the actual occurrence point k .

The core idea of BRR mechanism demonstrated by Alg. 2, is to dynamically adjust the weight distribution under the

Algorithm 2 Bipartite Randomized Response (BRR)

Input: quality loss matrix $\{\lambda_j^{(k)}\}_{N \times N}$ with $0 \leq \lambda_1^{(k)} \leq \lambda_2^{(k)} \leq \dots \leq \lambda_N^{(k)}$, privacy budget ϵ

- 1: Initialize: $s_1^{(k)} = e^\epsilon, s_2^{(k)} = \dots = s_N^{(k)} = 1, m^{(k)} = 1$
- 2: **for** $k = 1$ **to** N **do**
- 3: $w_1 = \frac{e^\epsilon}{e^\epsilon + N - 1}, w_2 = \dots = w_N = \frac{1}{e^\epsilon + N - 1}, Q^{(k)} = \sum_i \lambda_i^{(k)} w_i$
- 4: Search locally optimal $m^{(k)}$ of highest-utility items by Alg. 1
- 5: **end for**
- 6: Compute $m = \min_k m^{(k)}$
- 7: Update all weights w_j for $j = 1$ **to** N by

$$w_j = \begin{cases} \frac{e^\epsilon}{me^\epsilon + N - m}, & \text{if } j \leq m \\ \frac{1}{me^\epsilon + N - m}, & \text{if } j > m \end{cases}$$

Output: $(w_1, \dots, w_N)$ and m

	λ_1	λ_2	λ_3	λ_4	λ_5	λ_6	λ_7	λ_8	λ_9	λ_{10}	λ_{11}	λ_{12}		λ_N
1	0	1	2	3	4	5	6	7	8	9	10	11		N-1
2	0	1	1	2	3	4	5	6	7	8	9	10		N-2
3	0	1	1	2	2	3	4	5	6	7	8	9		N-3
4	0	1	1	2	2	3	3	4	5	6	7	8		N-4
5	0	1	1	2	2	3	3	4	4	5	6	7		N-5
6	0	1	1	2	2	3	3	4	4	5	5	6		N-6
7	0	1	1	2	2	3	3	4	4	5	5	6		N-7
⋮														⋮
N/2	0	1	1	2	2	3	3	4	4	5	5	6		N/2

Figure 3: Monotonicity characteristics of BRR with Euclidean distance utility for the array from 1 to even N . The symbol $l \prec l'$ in the column labeled by λ_i means that the partial derivative $\frac{\partial Q}{\partial s_i}$ for the row of l' is higher than that for l .

constraints of privacy protection, i.e., achieving an optimal trade-off between utility and privacy. In essence, the algorithms above aims to give the number m of high weighted candidates. Next, we present some explicit formulas of the uniform number m .

3.4 Explicit Formulas for m and Utility

As above in Section 3, the utility λ is denoted by the Euclidean distance between the actual and reported numbers, i.e., simply the absolute value of their difference $|x - y|$.

As illustrated in Fig. 3, after sorting λ of all candidates for reporting, we have $0 = \lambda_1 \leq \lambda_2 \leq \dots \leq \lambda_N \leq N$. For the numbers $1 \sim N$, the distance array $[\lambda]$ for item 1 is $[0, 1, 2, 3, \dots, N-1]$; the distance array for item 2 is $[0, 1, 1, 2, 3, \dots, N-2]$. For even N , we need only to consider the priori $k \leq N/2$ due to the symmetry.

Since the distances are sorted in an ascending order, for

Algorithm 3 BRR in one-dimensional equidistant point domain

Input: Privacy budget ϵ , equidistant points count N

- 1: $m_1 \leftarrow \left\lfloor \frac{\sqrt{N^2 e^\epsilon + \frac{1}{4}(1-e^\epsilon)^2} - (N - \frac{e^\epsilon}{2} + \frac{1}{2})}{e^\epsilon - 1} \right\rfloor$
- 2: **if** $\epsilon \geq 1$, **then** $m = m_1$ and go to Line 6
- 3: **if** N is even **then** $i = \left\lfloor \frac{N}{e^{\frac{\epsilon}{2}} + 1} + 1 \right\rfloor$, **otherwise**,
 $i = \left\lfloor \frac{\sqrt{e^\epsilon(N^2 - 1) + 1} - N}{e^\epsilon - 1} + 1 \right\rfloor$
- 4: **if** i is odd **then** $m_2 = i$, **otherwise**, $m_2 = i + 1$
- 5: $m \leftarrow \min(m_1, m_2)$
- 6: Define all weights w_j for $j = 1$ to N by

$$w_j = \begin{cases} \frac{e^\epsilon}{me^\epsilon + N - m}, & \text{if } j \leq m \\ \frac{1}{me^\epsilon + N - m}, & \text{if } j > m \end{cases}$$

Output: $(w_1, \dots, w_N)$ and m

any adjacent priori numbers k and $k' = k - 1$, by comparing their sorted distance arrays (rows) $[\lambda]$ and $[\lambda']$, we can find that there exists a single j_0 , such that $\lambda'_1 = \lambda_1, \lambda'_2 = \lambda_2, \dots, \lambda'_{j_0} = \lambda_{j_0}$, while for the remaining $N - j_0$ distances, $\lambda'_{j_0+1} = \lambda_{j_0+1} + 1, \lambda'_{j_0+2} = \lambda_{j_0+2} + 1, \dots, \lambda'_N = \lambda_N + 1$. Each column number j_0 is indicated by the column involving a gray ellipse area between the two rows.

During the local search phase by Alg. 1, one can mainly mention the sign of the derivative $\frac{\partial Q}{\partial s_i}$ for search the optimal local m . For the symbol $l < l'$ in the column labeled by λ_i , their involved arrays are denoted by $[\lambda]$ and $[\lambda']$, respectively, and the inequality $\frac{\partial Q}{\partial s_l} \Big|_{[\lambda]} < \frac{\partial Q}{\partial s_{l'}} \Big|_{[\lambda']}$ holds true based on the setting $s_1 = \dots = s_{i-1} = e^\epsilon, s_{i+1} = \dots = s_N = 1$. This indicates that the array $[\lambda']$ achieves the non-negative sign on the partial derivatives no later than array $[\lambda]$ and their locally optimal number of high-weighted candidates $m' \leq m$. This claim will be discussed in the proof of Theorem 1.

For the detailed explicit formulas on global m , we collect them regularly in Algorithm 3. Due to the inequalities marked by $<$ between the partial derivatives at the adjacent elements in the same column, we derive that the global minimal m can occur only in the row with the extreme or middle priori point. By comparing the two candidates, we prove that when the privacy budget $\epsilon \geq 1$, the first row (from Fig. 3) for the extreme points, 1 and N (each as priori point), give the global m for BRR (Lines 2-3). On the other hand, the last row (from Fig. 3) for the middle priori points, the explicit formulas are presented on Lines 4-5.

Following this, we obtain a series of main results as follows.

First, for the BRR mechanism in one-dimensional equidistant-point domain, the explicit formulas are mainly included in Theorem 1, and more detailed formulas are pre-

sented by Algorithm 3.

Theorem 1. Given privacy budget ϵ , let m be the number of items endowed with high probability in BRR mechanism defined by (10).

1) Whatever the selections of m_1 and m_2 , we have $m \geq 1$ and $\lim_{N \rightarrow \infty} \frac{m}{N} = \frac{1}{e^{\frac{\epsilon}{2}} + 1}$.

2) Assume $\epsilon \geq 1$, then we have, for the BRR mechanism,

$$m = \left\lfloor \frac{\sqrt{N^2 e^\epsilon + \frac{1}{4}(1-e^\epsilon)^2} - (N - \frac{e^\epsilon}{2} + \frac{1}{2})}{e^\epsilon - 1} \right\rfloor.$$

3) BRR satisfies ϵ -local differential privacy.

The proof is provided in Appendix A.1.

Afterwards, for the quality loss of BRR in equidistant point domain, we confirm theoretically that BRR outperforms GRR and obtain the exact upper and lower asymptotic bounds on the ratio for local quality loss of BRR over N at any given priori number k as well as the global quality loss ratio in Theorem 2.

Theorem 2. For sufficiently large count N , the asymptotic upper and lower bounds for the local expected errors of BRR over N are $\limsup_{N \rightarrow \infty} \frac{Q^{(k)}(BRR, N)}{N} = \frac{1}{e^{\frac{\epsilon}{2}} + 1}$ and

$$\liminf_{N \rightarrow \infty} \frac{Q^{(k)}(BRR, N)}{N} = \frac{1}{2(e^{\frac{\epsilon}{2}} + 1)}, \text{ respectively.}$$

The assertion follows by exploiting the symmetry of the equidistant domain. We consider the number n of pairs due to their distance to the priori position and compare $2n$ with the number m of items endowed with high probability. The detailed proof is provided in Appendix A.2.

Compared to the above bounds, those for GRR are $1/2$ and $1/4$, respectively which are larger correspondingly. These also give the upper and lower bounds of the ratio $Q_g(BRR)/N$ for global expected errors. The gap between the bounds exists obviously due to the pointwise estimate of the local ratios. In order to close this gap, the asymptotically exact ratio is presented as follows.

Theorem 3. For sufficiently large count N in the equidistant point domain, the asymptotically exact ratio for global expected error of BRR over N is

$$\lim_{N \rightarrow \infty} \frac{Q_g(BRR, N)}{N} = \frac{7e^{\frac{\epsilon}{2}} + 9}{12(e^{\frac{\epsilon}{2}} + 1)^2}.$$

This statement is obtained by integrating the local expected error computed above and then substituting the asymptotic ratio c by c_0 in the global expected-error expression as N tends to infinity. The complete proof is given in Appendix A.3.

The above assertions show that the global expected-error ratio for BRR is never larger than $1/3$ that is the limiting ratio for GRR, even at up to 66.4% reduction for $\epsilon = 3$.

Now we are ready to close the five challenges presented in Section 2.7 as follows.

- In the local search phase, we increase the prob. weights in sequence of utility. When the weight of some point ascends, the (normalized) weighted average error of reporting the other $N - 1$ points is fixed. Then, the monotonicity of the utility function for varying the weight of only one point can easily deduce the bipartite distribution of weight values, only 1 and e^ϵ .
- In the global search phase, we set the minimal $m^{(i)}$ obtained in the local search phase as the uniform number m of high probability weight e^ϵ , which is certainly independent of the priori distribution. Indeed, BRR assigns the minimal $m^{(i)}$ among all priori points, $i = 1, \dots, N$, as the final m . This value corresponds to the priori point, starting with which the BRR first achieves its own smallest (local) expected error when simultaneously increasing the number of high probability e^ϵ -weighted candidates for starting priori points.
- We present how to compute the uniform number m using point-wise derivative formulas in the general case, as shown by Alg. 2. In the equidistant point domain, the explicit formulas are simply presented by Alg. 3.
- The impacts are amazing in theoretic and application aspects. Theoretically, the local search algorithm begins with the GRR, and the global setting $m = \min_i m^{(i)}$ ensures that the BRR outperforms the GRR if $m > 1$ and otherwise equals. In the equidistant point domain some significant theoretical results are presented in Theorems 1 to 3 as demonstrated in Fig. 5. In applications to engineering including decision tree training, stochastic gradient descent, location-based services, and vector perturbation, a series of experimental results are shown in Section 5.
- For the frequency estimation scenario, BRR directly degenerates to be the same as GRR. Distribution estimation is an essential server-side post-processing step in LDP data collection in which GRR is a primary obfuscation mechanism. There is the risk that the not-uniform probability distribution introduced by BRR increases the exposure to poisoning. We consider to employ EMS (Expectation Maximization with Smoothing) [19], a likelihood-based estimator that iteratively infers the underlying frequency distribution from aggregated privatized reports while ensuring that the estimates remain smooth. We compare BRR with GRR under two baseline poisoning attacks, RIA and ROA, to rank estimation. We apply EMS estimator to isolate the benefit of BRR's adaptive, utility-aware probability allocation in Section 5.2.

4 Exploring the scenarios of BRR

In this section, we explore the scenario of BRR mechanism for various applications.

Algorithm 4 BRR in one-dimensional interval range

Input: Privacy budget ϵ , interval $[a, b]$, equidistant points count N , actual query value x

- 1: Generate set $\mathcal{Y}$ consisting of $y_i = a + \frac{j-1}{N-1}(b-a)$ for $j = 1, \dots, N$
- 2: Find the point y^* closest to x by (15)
- 3: Confirm the number m of high-weighted points by Algorithm 3
- 4: Construct the set Y_m by the m closest points to y^* in $\mathcal{Y}$
- 5: Generate a reported value y_j in $\mathcal{Y}$ with the distribution by (10)

Output: m and reported value y_j

4.1 Adapting BRR in Continuous Domain

We first consider the scalar case of continuous domain. That is, the perturbation interval defined as the scalar range of query function is equally divided, which presents a discrete domain consisting of N discrete nodes,

$$\mathcal{Y} = \{y_1, y_2, \dots, y_N\}.$$

Then, as shown in Algorithm 4, the actual query value x is firstly mapped to the closest discrete value $y^* \in \mathcal{Y}$:

$$y^* = \arg \min_{y_i \in \mathcal{Y}} |x - y_i|. \quad (15)$$

Afterwards, we make a bijection mapping the N equidistant points to their labels $1, 2, \dots, N$ in order. The linear transformation helps to present the BRR mechanism on the domain of $\mathcal{Y}$. The BRR perturbs the approximate value y^* to generate a reporting value y_j . Since the BRR randomized response mechanism provides LDP guarantees, this method can flexibly improve the discretization accuracy by increasing the number of nodes while enforcing ϵ -LDP.

The input of BRR already contains a certain level of noise during the mapping to discrete points. Specifically, since the original query value x is mapped to the closest point y^* in the discrete set $\mathcal{Y}$, this mapping process introduces a deviation from the original x , causing data distortion. Therefore, to ensure the approximation accuracy, we can reduce the mapping error by increasing the number of discrete points N . As N increases, the smaller intervals L among the discrete points become finer, making the mapped value y^* closer to the original data x , thus maintaining higher data accuracy.

However, we should note that increasing N , while reducing mapping errors, also increases the number of discrete points that need to be handled in the subsequent BRR or other mechanisms, thus expanding the spatial domain. This can lead to a sparser probability distribution in the release, potentially affecting the accuracy of the perturbation process. Therefore, although increasing N can improve the approximation of the mapping, it may destroy data utility. This violation really takes place for GRR, geometric and exponential mechanisms

but not for BRR due to its dynamic number selection of high probability candidates. This will be demonstrated in Section 5. For the vector case of query range, we can also develop the discrete domain in each dimension of the query range. More discussions will be demonstrated in Section 5.4.

4.2 BRR in General Discrete Domain

In this section, we consider to explore the scenario of BRR mechanism from two aspects: the distribution of points in the reporting domain, and the metric of data utility.

For stochastic distribution in the multi-dimensional discrete domain, or general metrics of data utility, one can perform the search in terms of Algorithm 2 with necessary adjustments on the ranking of λ_i and the sign condition of partial derivatives of utility. Specifically, for any apriori point x_k in the domain, the point x_k itself has the highest utility $\lambda_1^{(k)}$ and the non-increasing sequence $\lambda_1^{(k)} \geq \lambda_2^{(k)} \geq \dots \geq \lambda_N^{(k)}$, which presents the input of utility matrix $\{\lambda_j^{(k)}\}_{N \times N}$. Following this, we observe the partial derivative of utility and execute (Line 6):

if $\frac{\partial Q^{(k)}}{\partial s_i^{(k)}} > 0$, then $s_i^{(k)} = e^\epsilon$, $m^{(k)} = i$, otherwise, break.

Due to the intrinsic design of BRR, we can make the claim on the comparison to GRR as follows.

Theorem 4. *Given general discrete domain and utility function, for any priori point in the domain, BRR at least equals GRR on local utility. In addition, BRR outperforms GRR on global utility if $m > 1$.*

We want to mention that in the two-dimensional domain case, the BRR mechanism seems to be a simplification of the L-SRR mechanism [29], where instead of having multiple steps in the staircase, there are only two steps. That is not the case. In fact, as claimed above, developing BRR is motivated by promoting SOTA methods (e.g., L-SRR) for data release. The innovation starts exactly from local optimal search as shown in Alg. 1. We prove strictly that the two steps of probability weights give the optimal utility in the local search phase. The global search phase assigns the minimum $m^{(k)}$ as the uniform number m independent of prior distribution. Our proposed BRR is fit for LDP applications in any discrete domain (e.g., LBS scenarios) as optimal promotion of SOTA release methods while L-SRR mechanism can only be used in two-dimensional discrete domains due to its involved hierarchical encoding scheme.

5 Experimental Evaluations

In this section, the experimental evaluations consist of three parts:

- **Evaluations in discrete domains** (Sections 5.1 and 5.3). We first consider the basic scenario of equidistant point

domains as $\{1, 2, \dots, N\}$ for the alternative to GRR [15], Geometric [11] and exponential mechanism [22] by BRR, assigning Euclidean distance. The case of Jaccard similarity as utility is discussed in Appendix B.1 Afterwards, we illustrate the effect of BRR in 2-dimensional discrete domains for LBSs by comparing with GRR and the state-of-the-art L-SRR [29]. The runtime comparisons are presented in Appendix B.4.

- **Robustness evaluations against ranking-estimation poisoning attacks** (Section 5.2). We further investigate the robustness of BRR and GRR for ranking estimation under RIA and ROA [39] poisoning attacks.
- **Applications in continuous domains for machine learning** (Section 5.4 and Appendix B.2 to B.3). We evaluate the effect of applying BRR to machine learning (ML) for centralized DP, replacing Laplace mechanism in private GBDT [18], SGD [1], and vector perturbation for DNN, cf. [33].

These evaluations would confirm the robustness of BRR in a series of settings, particularly across multiple branches in machine learning, and illustrate that the effects of BRR are extensive and overwhelming in various scenarios of LBS tasks and applications.

5.1 Quality loss comparisons to GRR, Geometric and Exponential mechanisms

As we know, the data utility is negatively related to the distance, like Euclidean and Jaccard distances. Following Section 3, we define Euclidean distance as the data utility (loss) λ here. Then we conduct the experiments in the equidistant point domain $\mathcal{X} = \{1, 2, \dots, N\}$ with varying N . The standard metrics for comparisons among BRR, GRR and Exponential mechanisms include Q_g ratios and $QLoss$, where

$$QLoss = \frac{Q_g}{N-1} = \frac{\sum_{k=1}^N \sum_{i=1}^N \lambda_i^{(k)} w_i}{N(N-1)}, \quad (16)$$

with the diameter normalization of domain $\mathcal{X}$.

Fig. 4 shows that, under the same privacy budget, the quality loss of BRR is consistently lower than that of the GRR, Geometric and Exponential mechanism. This indicates that the BRR mechanism releases more effective candidates with higher probability given the privacy budget. We also observe, from Fig. 5(a), that as N increases, the gap in utility loss under the GRR mechanism grows significantly, whereas the utility loss of the BRR and exponential mechanisms remains relatively stable across different values of N . The reason is that only the true value is assigned a high weight of e^ϵ in the GRR mechanism while the remaining $N-1$ values are assigned weight 1, leading to increasing sparsity and dispersion of the probability distribution as N grows. In contrast, the

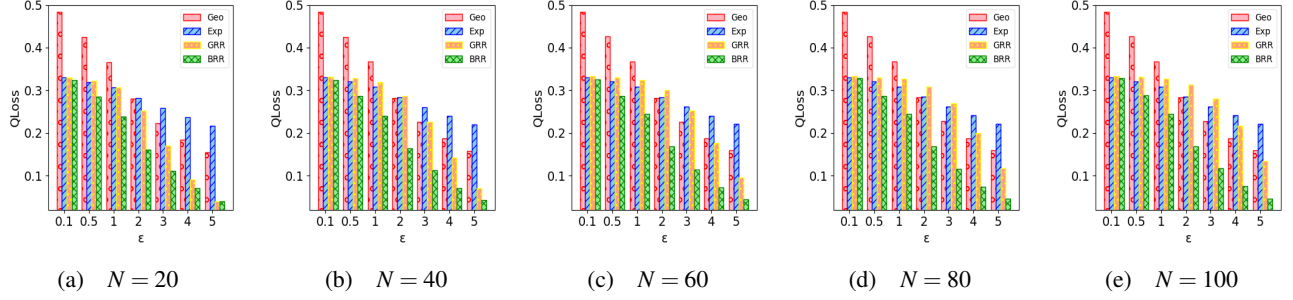


Figure 4: Quality loss comparison of BRR, GRR, Geometric (Geo) and Exponential (Exp) mechanisms using Euclidean distance with varying privacy budgets and count of candidates.

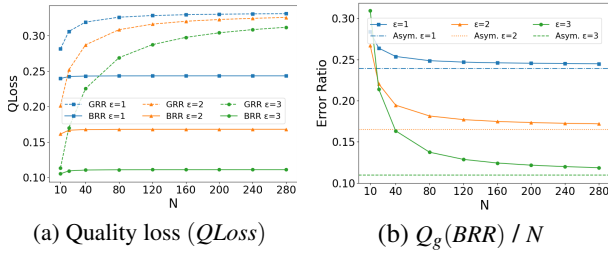


Figure 5: Comparison of quality loss in the equidistant point domain: (a) Quality loss ($QLoss$) of BRR and GRR mechanisms with varying total number N of priori points. (b) Ratio for global expected error of BRR over N as N varies. The asymptotically exact ratios (“Asym.”) are plotted, cf. Theorem 3.

BRR mechanism assigns the high weight e^ϵ to m values, and m increases with a asymptotic proportion to N , thus maintaining a more stable probability of effective responses, as shown in Fig. 5(a). The Geometric and Exponential mechanism exhibits similar behavior, effectively mitigating the impact of distribution sparsity.

Furthermore, as demonstrated by Fig. 5(b), when N tends to infinity, the asymptotically exact ratio for the quality-loss (and global expected-error) of BRR over N verifies the assertions given by Theorem 3 and decreases quickly as the privacy budget grows, e.g., approximately at the reduction of 49.3% and 66.4% for $\epsilon = 2$ and 3, respectively, compared to GRR. The superior performance of the BRR in maintaining data utility indicates its efficacy and potential for practical applications in privacy-preserving data releases.

As shown in Fig. 5(b), Fig. 4 demonstrates that the larger ϵ gives the smaller $QLoss$ (or Q_g) ratio of BRR over three methods that means greater improvements. However, this vigorous trend vanishes in the later tasks and applications. Particularly, for machine learning the obfuscation mechanisms are adopted as a part of the whole machine learning framework with centralized DP. The larger ϵ gives the lower privacy level,

the accuracy of iterative learning tasks would naturally approach that of non-private learning, and the smaller gaps on performances provide less chances to close.

5.2 Robustness against Ranking-estimation Poisoning Attacks

We evaluate the robustness of BRR and GRR against ranking-estimation poisoning attacks under normal and exponential data distributions. We pick the two least frequent items as attack targets. Since the target initially has the lowest frequency, a smaller post-attack rank indicates a stronger promotion effect. We measure the ranking change by

$$\Delta \text{Rank} = |\text{Rank}_{\text{poisoned}} - \text{Rank}_{\text{clean}}|. \quad (17)$$

The smaller ΔRank indicates more stable ranking estimate and stronger resistance to poisoning.

The total number of users is $n = 50,000$, and the domain size is $N = 60$. The server applies the EMS algorithm [19] to estimate item frequencies and construct the ranking. We consider two control knobs: (1) the poisoning ratio is fixed at $r = 0.1$ while the privacy budget varies over $\epsilon \in \{1, 2, 3, 4, 5, 6\}$; (2) the privacy budget is fixed at $\epsilon = 1.0$ while the fake-user ratio varies over $r \in \{0.01, 0.03, 0.05, 0.07, 0.10, 0.15\}$. Each configuration is repeated ten times, and the curves report the average results. In the RIA, according to the target items we set, each fake user uniformly and randomly picks a target item, perturbs this value using the LDP protocol, and then reports the perturbed value to the server [39]. In the ROA, after LDP perturbation, each fake user uniformly and randomly selects a target item and uses this item itself as the perturbed value [39]. Detailed results are shown in Fig. 6 and 7.

Randomized Input Attack (RIA). Under RIA, BRR generally produces less variable ΔRank than GRR for both distributions. As privacy budget ϵ increases, the ranking changes of both mechanisms become more sensitive to the injected inputs, while BRR mutates less, with the average ΔRank gap in a normal distribution reaching up to 30.15%. When the poisoning ratio increases, BRR shows a gradual increase in

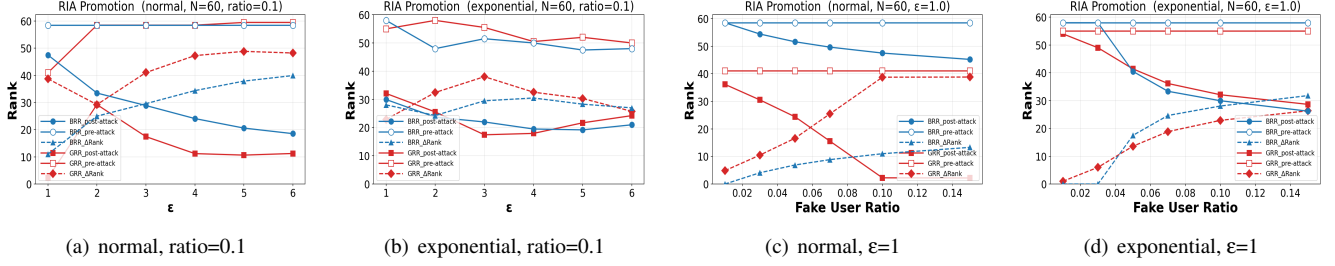


Figure 6: Comparison of the average rank of the target item for ranking estimation between BRR and GRR under RIA poisoning attacks over two data distributions, with varying privacy budget ϵ and fake user ratio.

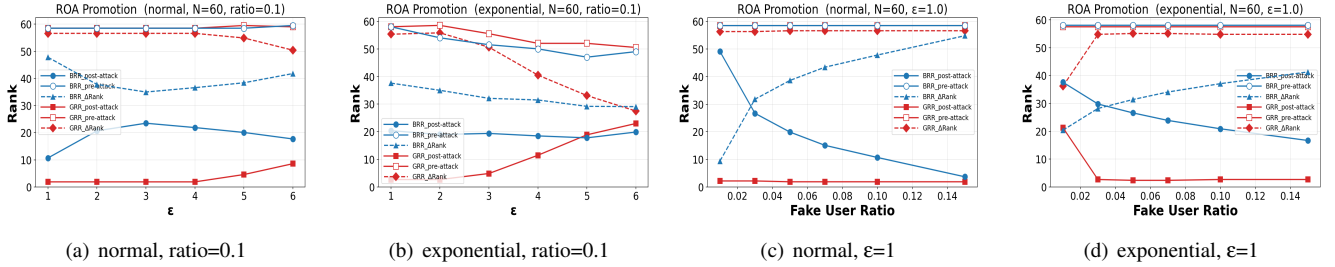


Figure 7: Comparison of the average rank of the target item for ranking estimation between BRR and GRR under ROA poisoning attacks over two data distributions, with varying privacy budget ϵ and fake user ratio.

Δ Rank, while GRR may promote the target more aggressively at the cost of a larger ranking displacement. Under the normal distribution, the average gap of Δ Rank can reach up to 70.32%. These results indicate that BRR provides more stable protection against RIA.

Randomized Output Attack (ROA). ROA directly injects fake outputs after perturbation and therefore generally constitutes a stronger attack surface than RIA. Under ROA, BRR still maintains a more smoother ranking response, with a smaller Δ Rank than GRR in most settings. For the normal distribution scenario, the average Δ Rank gap reaches 28.23% when varying ϵ , and 39.49% when varying the poisoning ratio. As ϵ or the poisoning ratio changes, GRR can achieve a faster target promotion but also causes larger fluctuations in the estimated ranking. In contrast, BRR better constrains rank displacement and preserves the overall ranking structure.

Overall, ROA exerts a stronger influence on the recovered ranking than RIA because it bypasses the input-side perturbation. Nevertheless, with identical attack type and parameter setting, BRR consistently provides a more stable ranking response than GRR, shown by its smaller Δ Rank.

5.3 BRR in Location-based Services

In location-based services (LBS), ensuring user privacy while maintaining high utility is a significant challenge. The L-SRR framework proposed by Wang et al. [29] employs a staircase randomized response mechanism to enhance utility while

achieving provable LDP. In this part, we make evaluations of BRR mechanism in the LBS scenario with comparison to the state-of-the-art L-SRR.

We conduct experiments on Gowalla² and FourSquare [36] datasets. We extract randomly 1,000 locations from the Gowalla dataset and 3,000 locations from the FourSquare dataset. To ensure consistency and facilitate comparison, following the hierarchical coding approach of the L-SRR scheme, we partition the datasets into regions based on common prefixes as shown in Fig. 8. These three obfuscation mechanisms share the same regional partitioning for release. The release range originating from a prior location is identical to its located region, which prevents cross-region releases.

We compare the performance of three schemes (L-SRR, GRR, and BRR) under different privacy budgets (ϵ ranging from 0.25 to 2). As shown in Fig. 9, we can observe that BRR performs the best across all privacy budgets. On the Gowalla dataset, the average global expected error Q_g of the BRR is reduced by 8.7% than that of L-SRR, at the most significant reduction up to 23.4% when $\epsilon = 2.0$. Similarly, on the FourSquare dataset, the average Q_g of BRR decreases by 6.7%, with the highest improvement by 14.3% at $\epsilon = 0.25$. Overall, when the privacy budget is low, the data utility of all schemes is significantly affected, but BRR still outperforms the other schemes, demonstrating its better noise resistance. As the privacy budget increases, the overall utility loss

²<https://snap.stanford.edu/data/loc-Gowalla.html>

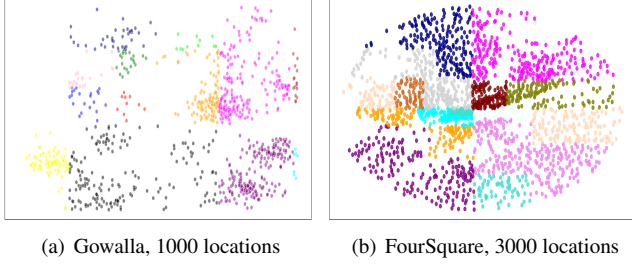


Figure 8: Two dataset maps are divided by the hierarchical coding approach involved in L-SRR scheme [29] into 13 and 14 regions, respectively, marked in colors. The smallest region contains 7 and 64 locations, respectively, while the largest region contains 235 and 278 locations, respectively.

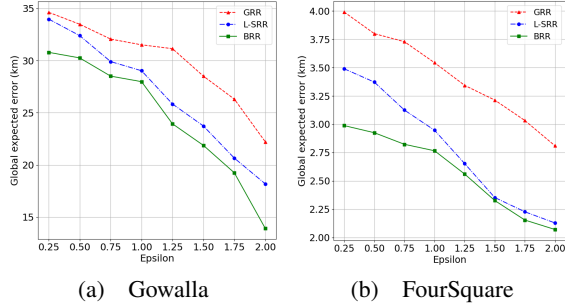


Figure 9: Global expected errors of L-SRR [29], GRR [15], and BRR with varying privacy budgets.

gradually decreases while BRR maintains relatively optimal accuracy across the entire range.

5.4 BRR for Vector Perturbation in DNN

To evaluate the effectiveness of the BRR for vector perturbation in ML, we adopted the DNN-DP framework proposed by Wang et al. [33] as a comparative baseline. To ensure a fair comparison, we replace the original Laplace noise by BRR mechanism while keeping the rest of DNN-DP unchanged. This allows us to compare the two mechanisms' abilities by uniform hyper-parameter optimization to preserve model's performance under privacy constraints.

The experiments are conducted on two publicly available datasets: US Census dataset³, and Bank Marketing dataset⁴ which contains records from a direct marketing campaign conducted by a Portuguese bank. The former contains 14 features that were used by [33] in the initial DNN-DP experiments, while the latter has been widely adopted in taxonomic studies and contains 16 features. In order to ensure the fairness and comparability of experimental results, we adopt a uniform

³<https://archive.ics.uci.edu/ml/datasets/Adult>

⁴<https://archive.ics.uci.edu/ml/datasets/Bank+Marketing>

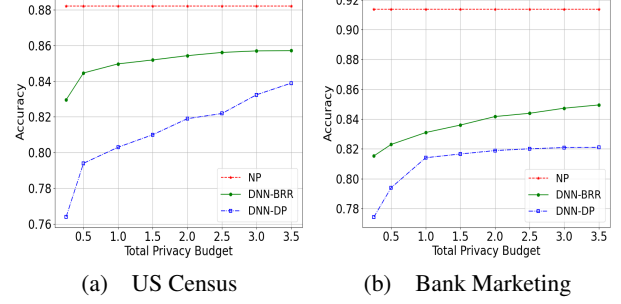


Figure 10: Classification-accuracy of DNN-DP approaches using the Laplace [33] and BRR mechanisms (DNN-BRR), respectively, and the vanilla DNN (NP), on the US Census and bank datasets, with varying total privacy budgets, cf. DNN-DP [33] (using Laplace), DNN-BRR (using BRR), NP [33].

hyperparameter optimization strategy. We vary the privacy budget within the range of 0.25 to 3.5 to systematically evaluate the classification accuracy of the two mechanisms under different levels of privacy protection.

As shown in Fig. 10, the experimental results on the US Census demonstrate that the BRR-based perturbation approach DNN-BRR consistently outperforms the traditional DNN-DP framework and also the results shown in [33] as the privacy budget increases from 0.25 to 3.5. On average, BRR achieves 3.22% improvement in accuracy over DNN, with a prominent gain of 5.29% at the lowest privacy budget ($\epsilon=0.25$). Similar results can be observed on the bank dataset where the BRR mechanism also exhibits superior performance compared to the Laplace-based DNN-DP method.

6 Related Work

This section mainly reviews the advancements in LDP mechanisms and recent applications in machine learning and LBS scenarios.

6.1 Advancements in LDP Mechanisms

Dwork et al. [8] first systematically introduced the concept of differential privacy, theorizing the obfuscation mechanism based on sensitivity. However, some data providers do not trust the data collectors and analysts. This induces researchers to study the setting of local privacy. Local privacy dates back to Warner [34], who designed in 1965 the randomized response method for individuals responding to sensitive surveys. Kasiviswanathan et al. [16] first formalized the idea of local privacy, and later Duchi et al. [7] studied the privacy-accuracy trade-offs in data tasks such as mean estimation, establishing a theoretical foundation for the application of LDP. These works rendered LDP prominent in the academic circle. Holohan et al. [13] provided an optimal mechanism for Warner's

original RR technique, achieving the best application of differential privacy in randomized response surveys.

With advances in technology, the k -ary randomized response [14] (k -RR, also known as GRR) extended traditional binary response to satisfy differential privacy in multi-value data scenarios. Building on GRR, Arcolezi et al. [2] introduced a series of enhanced LDP protocols, such as L-GRR, OUE (Optimized Unary Encoding), and SUE (Symmetric Unary Encoding), further improving the applicability of privacy protection in longitudinal data collection. Song et al. [26] further proposed the personalized randomized response (PRR) mechanism, enabling LDP to adapt more flexibly to individual differences by setting different privacy requirements for distinct data values. Wei et al. [35] proposed the advanced adaptive additive (AAA) mechanism, which is a distribution-aware approach that addresses the classical mean estimation problem. Wang et al. [30] proposed a generalizable aggregation framework to optimize protocol parameters through a simplified aggregation algorithm, improving the utility of data collection.

Although LDP protocols offer significant advantages in protecting user privacy, their distributed nature also makes it vulnerable to data poisoning attacks. Previous studies mainly investigate frequency estimation, whereas ranking attacks manipulate relative item frequencies to promote or demote target items. Zhan et al. [39] studied Randomized Input Attack (RIA) and Randomized Output Attack (ROA) for LDP ranking estimation.

6.2 Applications of LDP

LDP in Machine Learning. Through an in-depth analysis of existing studies, Wang et al. [32] and Chen et al. [4] summarized the applications and challenges of DP in deep learning and federated learning, respectively, including trade-offs between where noise is introduced, accuracy, and privacy, and the types of DP noise added. Bai et al. [3] combined differential privacy with decision tree models, proposing a multilayer DP model based on Markov Chain Monte Carlo to effectively balance privacy protection and model performance. Studies by Smith et al. [25] and Wang et al. [28] focused on optimizing interactive rounds and sample complexity of LDP, proposing methods to reduce computational costs and improve accuracy.

LDP in Distribution Estimation. Distribution estimation is an important server-side step in LDP data collection [30]. Previous methods mainly used unbiased estimation, such as empirical frequency estimation and matrix inversion, while producing infeasible results under strong noise. The Expectation–Maximization (EM) algorithm, originally proposed by Dempster et al. [6], provides a likelihood-based alternative by iteratively performing the expectation and maximization steps. For LDP distribution estimation, Li et al. [19] proposed EMS, which incorporates smoothing into EM to improve the stability of numerical distribution estimation.

Later, Ye et al. [38] introduced a mixture-reduction framework to alleviate the overfitting and computational costs of EM-based estimation. For randomized response, Pinzón et al. [24] derived an exact maximum-likelihood estimator without iterative Bayesian updating.

LDP in Location-based Services. Wang et al. [29] proposed a staircase randomized response mechanism named L-SRR that introduced the hierarchical coding scheme. This makes L-SRR suitable only in two-dimensional domain. By incorporating expected inference error and geo-indistinguishability, a regionalized LDP mechanism called DPIVE with personalized utility sensitivities was developed for LBSs [40]. Zhu et al. [41] introduced termed location-discriminative geo-indistinguishability involving different sensitive levels of location privacy.

LDP in other scenarios. Gursoy et al. [12] proposed the Condensed Local Differential Privacy (CLDP) mechanism, which focuses privacy protection on data collection for smaller groups, improving applicability in network security for small-scale data.

Kairouz et al. [14] presented new mechanisms, including hashed k -RR, which outperform existing mechanisms in utility across all privacy levels. Murakami and Kawamoto [23] introduced Utility-optimized LDP to improve utility in contexts containing large amounts of non-sensitive data.

To solve private release problems of multiple candidates, SRR partitions locations into multiple groups for staircase probabilities. However, hierarchical coding scheme is integrated in L-SRR to realize SRR for LBS which weakens the data utility. This motivates us to design an adaptive Bipartite Randomized Response mechanism called BRR in this paper which improves utility as desired.

7 Conclusion

This paper proposes a novel adaptive LDP release mechanism, the Bipartite Randomized Response (BRR), which aims to balance the trade-off between privacy and data utility. In particular, SRR partitions all locations into multiple groups for staircase probabilities which weakens the data utility. To address this limitation, BRR optimizes the allocation of release probabilities by local and global search phases. This optimization enhances data utility while preserving privacy level. We provide a series of theoretical results for the optimal number m of high-weighted items and asymptotically exact ratio for global expected errors of BRR over N in the equidistant point domains. Various empirical evaluations verify that our BRR approach outperforms the state-of-the-art methods while providing strong robustness against poisoning attacks.

For the future work, we will investigate some reformed Bipartite Random Response mechanisms for minimizing the global (unconditional) expected error in general discrete domains under uniform and prior distributions for various application scenarios.

Ethical Considerations

This work studies the design and evaluation of the Bipartite Randomized Response (BRR) mechanism for local differential privacy. The primary goal is to improve the utility of privacy-preserving data release while maintaining the privacy guarantees of LDP. Our experiments do not collect new personal data or involve human subjects. We use publicly available datasets and synthetic distributions only for evaluating the utility, ranking estimation, and robustness of the mechanisms. We do not attempt to identify, contact, or infer the identities of any individuals in the datasets.

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Open Science

Artifacts provided. To support the reproducibility of our experimental evaluation of the Bipartite Randomized Response (BRR) mechanism and its integration with other differential privacy (DP) techniques, we provide the following artifacts:

- Source code implementing the BRR mechanism (BRR/BRR.py) and the local utility optimization algorithm (BRR/LocalRS.py)
- Implementation of related schemes for Private GBDT in Appendix B.2 (DPBoost/)
- Reproduction code for the L-SRR scheme by Wang et al. [29] (L-SRR/encode.py, L-SRR/partition.py)
- Reproduction code for the EMS-based frequency estimation [19] and the ranking-estimation poisoning attacks, including the corresponding experimental scripts in Section 5.2 (DistributionEstimation/)
- Implementations of related schemes applied to Private SGD in Appendix B.3 and Vector Perturbation in DNN in Section 5.4 (SGDandDNN/)
- All experimental datasets are included in the repository, organized within each module’s directory (e.g., DPBoost/data for regression experiments)
- Documentation in the form of inline comments and this README, specifying input/output formats, parameter configurations, and execution instructions

Access during double-blind review. All artifacts are hosted in an anonymous, publicly accessible repository:

<https://anonymous.4open.science/r/BipartiteRR-0FC9>.

The repository is self-contained and can be executed independently in each subfolder. We recommend Python 3.12 and standard scientific libraries (e.g., NumPy, Scikit-learn, PyTorch). No authentication or special permissions are required.

Justification for withheld artifacts. All core artifacts necessary to reproduce the results presented in the paper are included in the repository. The experimental datasets are derived from publicly available sources, and no proprietary or sensitive data is used. Therefore, no materials are withheld.

References

- [1] Martin Abadi, Andy Chu, Ian Goodfellow, H Brendan McMahan, Ilya Mironov, Kunal Talwar, and Li Zhang. Deep learning with differential privacy. In *Proceedings of the 2016 ACM SIGSAC Conference on Computer and Communications Security*, pages 308–318, 2016.
- [2] Héber H. Arcolezi, Jean François Couchot, Bechara Al Bouna, and Xiaokui Xiao. Improving the utility of locally differentially private protocols for longitudinal and multidimensional frequency estimates. *Digital Communications and Networks*, 10(2):369–379, 2024.
- [3] Xuanyu Bai, Jianguo Yao, Mingxuan Yuan, Ke Deng, Xike Xie, and Haibing Guan. Embedding differential privacy in decision tree algorithm with different depths. *Science China Information Sciences*, 60:1–15, 2017.
- [4] Huiqiang Chen, Tianqing Zhu, Tao Zhang, Wanlei Zhou, and Philip S. Yu. Privacy and fairness in federated learning: On the perspective of tradeoff. *ACM Comput. Surv.*, 56(2), 2023.
- [5] Lea Demelius, Roman Kern, and Andreas Trügler. Recent Advances of Differential Privacy in Centralized Deep Learning: A Systematic Survey. *ACM Computing Surveys*, 57(6), 2025.
- [6] A. P. Dempster, N. M. Laird, and D. B. Rubin. Maximum likelihood from incomplete data via the em algorithm. *Journal of the Royal Statistical Society: Series B (Methodological)*, 39(1):1–22, 1977.
- [7] John C. Duchi, Michael I. Jordan, and Martin J. Wainwright. Local privacy and statistical minimax rates. In *2013 IEEE 54th Annual Symposium on Foundations of Computer Science*, pages 429–438. IEEE, 2013.
- [8] Cynthia Dwork, Frank McSherry, Kobbi Nissim, and Adam Smith. Calibrating noise to sensitivity in private data analysis. In *Theory of Cryptography: Third Theory of Cryptography Conference, TCC 2006, New York, NY, USA, March 4-7, 2006. Proceedings 3*, pages 265–284. Springer, 2006.
- [9] Úlfar Erlingsson, Vasyl Pihur, and Aleksandra Korolova. Rappor: Randomized aggregatable privacy-preserving ordinal response. In *Proceedings of the 2014 ACM SIGSAC Conference on Computer and Communications Security, CCS ’14*, page 1054–1067, New York, NY, USA, 2014. Association for Computing Machinery.
- [10] Huiyu Fang, Liquan Chen, Yali Liu, and Yuan Gao. Locally Differentially Private Frequency Estimation Based on Convolution Framework. In *2023 IEEE Symposium on Security and Privacy (S&P)*, pages 2208–2222. IEEE, 2023.
- [11] Arpita Ghosh, Tim Roughgarden, and Mukund Sundararajan. Universally utility-maximizing privacy mechanisms. In Michael Mitzenmacher, editor, *Proceedings of the 41st Annual ACM Symposium on Theory of Computing, STOC 2009, Bethesda, MD, USA, May 31 - June 2, 2009*, pages 351–360. ACM, 2009.
- [12] Mehmet Emre Gursoy, Acar Tamersoy, Stacey Truex, Wenqi Wei, and Ling Liu. Secure and utility-aware data collection with condensed local differential privacy. *IEEE Transactions on Dependable and Secure Computing*, 18(5):2365–2378, 2019.
- [13] Naoise Holohan, Douglas J. Leith, and Oliver Mason. Optimal Differentially Private Mechanisms for Randomised Response. *IEEE Transactions on Information Forensics and Security*, 12(11):2726–2735, 2017.
- [14] Peter Kairouz, Keith Bonawitz, and Daniel Ramage. Discrete distribution estimation under local privacy. In *International Conference on Machine Learning*, pages 2436–2444. PMLR, 2016.
- [15] Peter Kairouz, Sewoong Oh, and Pramod Viswanath. Extremal Mechanisms for Local Differential Privacy. *Journal of Machine Learning Research*, 17(17):1–51, 2016.
- [16] Shiva Prasad Kasiviswanathan, Homin K. Lee, Kobbi Nissim, Sofya Raskhodnikova, and Adam D. Smith. What Can We Learn Privately? *SIAM Journal on Computing*, 40(3):793–826, 2011.
- [17] Ninghui Li, Min Lyu, Dong Su, and Weining Yang. *Differential Privacy: From Theory to Practice*. Springer, 2017.
- [18] Qinbin Li, Zhaomin Wu, Zeyi Wen, and Bingsheng He. Privacy-preserving gradient boosting decision trees. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 34, pages 784–791, 2020.
- [19] Zitao Li, Tianhao Wang, Milan Lopuhaä-Zwakenberg, Ninghui Li, and Boris Škoric. Estimating numerical distributions under local differential privacy. In *Proceedings of the 2020 ACM SIGMOD International Conference on Management of Data, SIGMOD ’20*, page 621–635, New York, NY, USA, 2020. Association for Computing Machinery.
- [20] Yuheng Ma, Han Zhang, Yuchao Cai, and Hanfang Yang. Decision tree for locally private estimation with public data. *Advances in Neural Information Processing Systems*, 36:43676–43705, 2023.
- [21] Pathum Chamikara Mahawaga Arachchige, Peter Bertok, Ibrahim Khalil, Dongxi Liu, Seyit Camtepe, and Mohammed Atiquzzaman. Local differential privacy

- for deep learning. *IEEE Internet of Things Journal*, 7(7):5827–5842, 2020.
- [22] Frank McSherry and Kunal Talwar. Mechanism design via differential privacy. In *48th Annual IEEE Symposium on Foundations of Computer Science (FOCS'07)*, pages 94–103. IEEE, 2007.
 - [23] Takao Murakami and Yusuke Kawamoto. Utility-optimized local differential privacy mechanisms for distribution estimation. In *28th USENIX Security Symposium (USENIX Security 19)*, pages 1877–1894, 2019.
 - [24] Carlos Antonio Pinzón, Ehab ElSalamouny, Lucas Masot, Alexis Miller, Héber Hwang Arcolezi, and Catuscia Palamidessi. Estimating the true distribution of data collected with randomized response. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 40, pages 35751–35758, 2026.
 - [25] Adam Smith, Abhradeep Thakurta, and Jalaj Upadhyay. Is interaction necessary for distributed private learning? In *2017 IEEE Symposium on Security and Privacy (S&P)*, pages 58–77. IEEE, 2017.
 - [26] Haina Song, Tao Luo, Xun Wang, and Jianfeng Li. Multiple sensitive values-oriented personalized privacy preservation based on randomized response. *IEEE Transactions on Information Forensics and Security*, 15:2209–2224, 2020.
 - [27] Arnaud J. Tournier and Yves-Alexandre de Montjoye. Expanding the attack surface: Robust profiling attacks threaten the privacy of sparse behavioral data. *Science Advances*, 8(33), 2022. eabl6464.
 - [28] Di Wang, Marco Gaboardi, and Jinhui Xu. Empirical risk minimization in non-interactive local differential privacy revisited. *Advances in Neural Information Processing Systems*, 31, 2018.
 - [29] Han Wang, Hanbin Hong, Li Xiong, Zhan Qin, and Yuan Hong. L-SRR: Local differential privacy for location-based services with staircase randomized response. In *Proceedings of the 2022 ACM SIGSAC Conference on Computer and Communications Security*, pages 2809–2823, 2022.
 - [30] Tianhao Wang, Jeremiah Blocki, Ninghui Li, and Somesh Jha. Locally differentially private protocols for frequency estimation. In *26th USENIX Security Symposium (USENIX Security 17)*, pages 729–745, 2017.
 - [31] Tianhao Wang, Ninghui Li, and Somesh Jha. Locally differentially private heavy hitter identification. *IEEE Transactions on Dependable and Secure Computing*, 18(2):982–993, 2019.
 - [32] Yanling Wang, Qian Wang, Lingchen Zhao, and Cong Wang. Differential privacy in deep learning: Privacy and beyond. *Future Generation Computer Systems*, 148:408–424, 2023.
 - [33] Yufeng Wang, Min Gu, Jianhua Ma, and Qun Jin. Dnn-dp: Differential privacy enabled deep neural network learning framework for sensitive crowdsourcing data. *IEEE Transactions on Computational Social Systems*, 7(1):215–224, 2020.
 - [34] Stanley L. Warner. Randomized response: A survey technique for eliminating evasive answer bias. *Journal of the American Statistical Association*, 60(309):63–69, 1965.
 - [35] Fei Wei, Ergute Bao, Xiaokui Xiao, Yin Yang, and Bolin Ding. Aaa: An adaptive mechanism for locally differentially private mean estimation. *Proc. VLDB Endow.*, 17(8):1843–1855, Apr. 2024.
 - [36] Dingqi Yang, Bingqing Qu, Jie Yang, and Philippe Cudre-Mauroux. Revisiting user mobility and social relationships in lbsns: A hypergraph embedding approach. In *The World Wide Web Conference*, pages 2147–2157, 2019.
 - [37] Qingqing Ye, Haibo Hu, Xiaofeng Meng, and Huadi Zheng. Privkv: Key-value data collection with local differential privacy. In *2019 IEEE Symposium on Security and Privacy (S&P)*, pages 317–331. IEEE, 2019.
 - [38] Yutong Ye, Tianhao Wang, Min Zhang, and Dengguo Feng. Revisiting em-based estimation for locally differentially private protocols. *Proceedings 2025 Network and Distributed System Security Symposium*, 2025.
 - [39] Pei Zhan, Peng Tang, Yangzhuo Li, Puwen Wei, and Shanqing Guo. Poisoning attacks to local differential privacy for ranking estimation. In *Proceedings of the 2025 ACM SIGSAC Conference on Computer and Communications Security, CCS '25*, page 1215–1229, New York, NY, USA, 2025. Association for Computing Machinery.
 - [40] Shun Zhang, Pengfei Lan, Benfei Duan, Zhili Chen, Hong Zhong, and Neal N. Xiong. DPIVE: A regionalized location obfuscation scheme with personalized privacy levels. *ACM Transactions on Sensor Networks*, 20(2), 2024. Article 35, 26 pages.
 - [41] Youwen Zhu, Yuanyuan Hong, Qiao Xue, Xiao Lan, Yushu Zhang, and Yong Xiang. LDGI: Location-discriminative geo-indistinguishability for location privacy. *IEEE Transactions on Knowledge and Data Engineering*, 37(3):1282–1293, 2025.

A Proofs

A.1 Proof of Theorem 1

We begin with the illustration by Fig. 3. For each adjacent priori number k , the distances for all candidates are sorted in an ascending order. For any two adjacent rows with priori numbers k and $k' = k - 1$, respectively, we compare their sorted distance arrays (rows), $[\lambda]$ and $[\lambda']$. Then there exists a single j_0 , such that $\lambda'_1 = \lambda_1, \lambda'_2 = \lambda_2, \dots, \lambda'_{j_0} = \lambda_{j_0}$, while for the remaining $N - j_0$ items, $\lambda'_{j_0+1} = \lambda_{j_0+1} + 1, \lambda'_{j_0+2} = \lambda_{j_0+2} + 1, \dots, \lambda'_N = \lambda_N + 1$. Such a number j_0 is indicated by the column number involving a gray ellipse area between the two rows.

Now we consider to increase the probability weight s_i in order. We first consider the case $i \leq j_0$. When considering the weight increase on s_i , we have fixed $s_1 = \dots = s_{i-1} = e^\varepsilon$ for both rows, and for the numerator part of (12) about $\frac{\partial Q}{\partial s_i}$,

$$\begin{aligned} \sum_j (\lambda'_i - \lambda'_j) s_j &= \sum_{j=1}^{j_0} (\lambda'_i - \lambda'_j) s_j + \sum_{j=j_0+1}^N (\lambda'_i - \lambda'_j) s_j \\ &= \sum_{j=1}^{j_0} (\lambda_i - \lambda_j) s_j + \sum_{j=j_0+1}^N (\lambda_i - (\lambda_j + 1)) s_j \\ &= \sum_j (\lambda_i - \lambda_j) s_j - \sum_{j_0+1}^N s_j \leq \sum_j (\lambda_i - \lambda_j) s_j, \end{aligned}$$

where the equality on the last line holds only when $j_0 = N$. This shows the monotonicity relationship within the same column in the lower triangular red region, specifically indicating that the numerator factor of the partial derivative of utility loss for the item below is greater than the corresponding factor for the item above when considering increasing the probability weight.

Next, we consider the case $i > j_0$, then $\lambda'_i = \lambda_i + 1$. When $j \leq j_0$, $\lambda'_j = \lambda_j$; and when $j > j_0$, $\lambda'_j = \lambda_j + 1$. Then, for $i > j_0$, we have

$$\begin{aligned} \sum_j (\lambda'_i - \lambda'_j) s_j &= \sum_{j=1}^{j_0} (\lambda'_i - \lambda'_j) s_j + \sum_{j=j_0+1}^N (\lambda'_i - \lambda'_j) s_j \\ &= \sum_{j=1}^{j_0} (\lambda_i + 1 - \lambda_j) s_j + \sum_{j=j_0+1}^N (\lambda_i + 1 - (\lambda_j + 1)) s_j \\ &= \sum_{j=1}^{j_0} (\lambda_i - \lambda_j) s_j + \sum_{j=j_0+1}^N (\lambda_i - \lambda_j) s_j + \sum_{j=1}^{j_0} s_j \\ &= \sum_j (\lambda_i - \lambda_j) s_j + \sum_{j=1}^{j_0} s_j > \sum_j (\lambda_i - \lambda_j) s_j. \end{aligned}$$

This demonstrates the monotonicity relationship within the same column in the upper triangular green region, specifically indicating that the numerator factor of the partial derivative of utility loss for the item below is less than the corresponding factor for the item above.

Based on Fig. 3, if the threshold value i for changing the sign of the partial derivative appears before (no more than) j_0 (in the red region), the lower row will always reach the threshold before the upper one for any two adjacent rows. In this case, starting from the midpoint $N/2$ as the prior, the bottom-most row achieves the smallest m . On the other hand, if the threshold appears after (larger than) j_0 (in the green region), the upper row k will reach the threshold before the lower k' . In this scenario, starting from the extreme points 1 or N as the prior, the top-most row achieves the smallest m . Fig. 3 depicts the situation where N is even; the case for odd N is similar and will not be elaborated further.

Afterwards, it can be seen that the minimum m is obtained when the true item is at the extreme point or the middle point. If it is at the extreme point, we have

$$\begin{aligned} &\sum_{j=1}^{i-1} (\lambda_i - \lambda_j) e^\varepsilon + \sum_{j=i+1}^N (\lambda_i - \lambda_j) 1 \\ &= \frac{(1+i-1)(i-1)}{2} e^\varepsilon - \frac{(1+N-i)(N-i)}{2} \quad (18) \\ &= \frac{1}{2} (e^\varepsilon - 1) i^2 + \left(N - \frac{1}{2} e^\varepsilon + \frac{1}{2} \right) i - \frac{1}{2} N^2 - \frac{1}{2} N, \end{aligned}$$

which is a quadratic function in i , where $\Delta = N^2 e^\varepsilon + \frac{1}{4} (1 - e^\varepsilon)^2 > 0$. Since the integer i is positive, we get

$$m_1 = i = \left\lfloor \frac{\sqrt{N^2 e^\varepsilon + \frac{1}{4} (1 - e^\varepsilon)^2} - (N - \frac{e^\varepsilon}{2} + \frac{1}{2})}{e^\varepsilon - 1} \right\rfloor.$$

Since the arrangement of the median depends on N , if the minimum m is obtained at the middle point, it is necessary to consider the two different cases when N is odd or even. In the median, there are many paired values, making the paired i always odd. Here, only the case where i is odd needs to be considered.

For even N :

$$\begin{aligned} &\sum_{j=1}^{i-1} (\lambda_i - \lambda_j) e^\varepsilon + \sum_{j=i+1}^N (\lambda_i - \lambda_j) \times 1 \\ &= \frac{(e^\varepsilon - 1)}{4} i^2 + \frac{N - (e^\varepsilon - 1)}{2} i + \frac{(e^\varepsilon - 1) - N^2 - 2N}{4}, \quad (19) \end{aligned}$$

where $\Delta = \frac{e^\varepsilon}{4} N^2$. We have $i = \left\lfloor \frac{N}{e^{\frac{\varepsilon}{2}} + 1} + 1 \right\rfloor$.

For odd N :

$$\begin{aligned} &\sum_{j=1}^{i-1} (\lambda_i - \lambda_j) e^\varepsilon + \sum_{j=i+1}^N (\lambda_i - \lambda_j) \times 1 \\ &= \frac{(e^\varepsilon - 1)}{4} i^2 + \frac{N - (e^\varepsilon - 1)}{2} i + \frac{(e^\varepsilon - 1) - 2N - N^2 + 1}{4}, \quad (20) \end{aligned}$$

where $\Delta = \frac{e^\varepsilon}{4}(N^2 - 1) + \frac{1}{4}$. We have $i = \left\lfloor \frac{\sqrt{e^\varepsilon(N^2-1)+1}-N}{e^\varepsilon-1} + 1 \right\rfloor$. If i is odd, then $m_2 = i$; if i is even, then $m_2 = i + 1$.

So we just need to compare the m values obtained at the extreme point and the middle point, and take the smaller one as the final $m = \min\{m_1, m_2\}$.

For ease of discussion, let

$$z = \sum_{j=1}^{i-1} (\lambda_i - \lambda_j) e^\varepsilon + \sum_{j=i+1}^N (\lambda_i - \lambda_j) 1.$$

If the minimum m is obtained at the extreme point, we have

$$z_1(i) = \frac{(e^\varepsilon - 1)}{2} i^2 + \left(N - \frac{e^\varepsilon}{2} + \frac{1}{2} \right) i - \frac{N^2}{2} - \frac{N}{2}.$$

If the minimum m is obtained at the middle point, then for even N , we have

$$z_2(i) = \frac{(e^\varepsilon - 1)}{4} i^2 + \frac{N - (e^\varepsilon - 1)}{2} i + \frac{(e^\varepsilon - 1)}{4} - \frac{N}{2} - \frac{N^2}{4},$$

and for odd N , we have

$$z_3(i) = \frac{(e^\varepsilon - 1)}{4} i^2 + \frac{N - (e^\varepsilon - 1)}{2} i + \frac{(e^\varepsilon - 1)}{4} - \frac{N}{2} - \frac{N^2}{4} + \frac{1}{4}.$$

It can be observed that z_3 can be obtained by shifting z_2 upwards by $\frac{1}{4}$ units. Therefore, the left root of z_3 is greater than the left root of z_2 , and the right root of z_3 is smaller than the right root of z_2 . The left root of z_3 is $i_{3l} = 1 - \frac{N + \sqrt{e^\varepsilon(N^2-1)+1}}{e^\varepsilon-1} < 1$, thus the left root of z_2 is also less than 1.

Considering the selections of m_1 and m_2 , we have

$$\lim_{N \rightarrow \infty} \frac{m_1}{N} = \lim_{N \rightarrow \infty} \frac{m_2}{N} = \frac{1}{e^{\frac{\varepsilon}{2}} + 1}.$$

This gives the claim 1).

To analyze which of m_1 or m_2 is larger, it is necessary to compare the right root of z_1 , i_{1r} , with the right root of z_3 when N is odd, i_{3r} . We have first

$$\begin{aligned} i_{1r} &= \frac{\sqrt{N^2 e^\varepsilon + \frac{1}{4}(1 - e^\varepsilon)^2} - (N - \frac{e^\varepsilon}{2} + \frac{1}{2})}{e^\varepsilon - 1} \\ &< \frac{N e^{\frac{\varepsilon}{2}} + \frac{1 - e^\varepsilon}{2} - (N - \frac{e^\varepsilon}{2} + \frac{1}{2})}{e^\varepsilon - 1} = \frac{N e^{\frac{\varepsilon}{2}} - N}{e^\varepsilon - 1}, \\ i_{3r} &= \frac{\sqrt{e^\varepsilon(N^2 - 1) + 1} - (N - e^\varepsilon + 1)}{e^\varepsilon - 1} \\ &> \frac{e^{\frac{\varepsilon}{2}}(N - 1) - (N - e^\varepsilon + 1)}{e^\varepsilon - 1}. \end{aligned} \quad (21)$$

Let $a = N e^{\frac{\varepsilon}{2}} - N$, $b = e^{\frac{\varepsilon}{2}}(N - 1) - (N - e^\varepsilon + 1)$, then $b - a = e^\varepsilon - e^{\frac{\varepsilon}{2}} - 1 = (e^{\frac{\varepsilon}{2}} - \frac{1}{2})^2 - \frac{5}{4}$. Setting $b - a > 0$, we solve for $\varepsilon > 2 \ln \left(\frac{\sqrt{5}+1}{2} \right) \approx 0.9624$. Therefore, when $\varepsilon \geq 1$, we

always have $i_{3r} > i_{1r}$, and $m_1 = \lfloor i_{1r} \rfloor$, $m_2 = \lfloor i_{3r} \rfloor$, hence $m_1 \leq m_2$. This shows that when $\varepsilon \geq 1$, the minimum m is obtained at the extreme point. This gives the claim 2).

Given any released candidate $y \in \mathcal{Y}$, for any possible priori items $x_1, x_2 \in \mathcal{X}$, we have $\frac{\Pr[y|x_1]}{\Pr[y|x_2]} \leq e^\varepsilon$. This gives the claim 3).

A.2 Proof of Theorem 2

We only need to consider the case where N is a sufficiently large odd integer. Without loss of generality, let m also be an odd number satisfying $m \sim \frac{N}{e^{\varepsilon/2}+1}$. The symbol $\sim$ means the equivalence between two formulas for sufficiently large N .

Given any priori position $k \leq N$ with $k \in \mathbb{Z}^+$, the distances from any of N positions (to be possibly reported) to k can be arranged in ascending order. As illustrated in each row of Fig. 3, the general form of this ordered sequence is $[0, 1, 1, 2, 2, \dots, n, n, n+1, n+2, \dots, N-1-n]$, where $0 < 2n \leq N-1$ and $k = n+1$ or $N-n$. We now analyze the local expected errors $Q^{(k)}$ under two protocols.

For BRR mechanism, we distinguish three cases:

- (1) When $m-1 \leq 2n \leq N-1$, the $Q^{(k)}(BRR)$ is expressed as,

$$Q^{(k)}(BRR) = \frac{1}{N + (e^\varepsilon - 1)m} \cdot \left\{ \frac{m-1}{2} \left(\frac{m-1}{2} + 1 \right) e^\varepsilon + \left[\frac{(N-n)(N-n-1)}{2} + \frac{n(n+1)}{2} - \frac{m-1}{2} \frac{m+1}{2} \right] \right\}.$$

- (2) When $2n < m$, the $Q^{(k)}(BRR)$ becomes,

$$Q^{(k)}(BRR) = \frac{(m-1-n)(m-n) + n(n+1)}{2(N + (e^\varepsilon - 1)m)} e^\varepsilon + \frac{(N-n)(N-n-1) - (m-1-n)(m-n)}{2(N + (e^\varepsilon - 1)m)}.$$

- (3) When $n = 0$, the priori point is located at the extreme position $k = 1$ or N , and the sorted distance sequence is $[0, 1, 2, \dots, N-1]$, where 0 appears once. The $Q^{(k)}(BRR)$ is identical to the above formula with $n = 0$ in case (2).

In the following, we proceed with the computation employing asymptotic equivalent substitutions for sufficiently large N . We define the normalized parameters: $m = cN$, $n = dN$, with $c_0 = \frac{1}{e^{\varepsilon/2}+1}$, $c < \frac{1}{2}$ and $0 \leq d < \frac{1}{2}$.

For $Q^{(k)}(BRR)$, we discuss the asymptotic equivalent substitution of order in two cases for sufficiently large N .

- (1) Let's consider the case $m-1 \leq 2n \leq N-1$. Then,

$$\begin{aligned} Q^{(k)}(BRR)/N &= \frac{c}{2} \cdot \frac{c}{2} \cdot \frac{e^\varepsilon}{1 + (e^\varepsilon - 1)c} \\ &+ \left[\frac{(1-d)^2}{2} + \frac{d^2}{2} - \frac{c^2}{4} \right] \cdot \frac{1}{1 + (e^\varepsilon - 1)c} \\ &= \frac{1}{4(1 + (e^\varepsilon - 1)c)} [c^2(e^\varepsilon - 1) + 2(1 - 2d + 2d^2)]. \end{aligned}$$

Considering the inequality $cN - 1 \leq 2dN \leq N - 1$, we have $\frac{c}{2} \leq d < \frac{1}{2}$ for sufficiently large N . Moreover, $\frac{1}{2} \leq 1 - 2d + 2d^2 = 2(d - \frac{1}{2})^2 + \frac{1}{2} \leq 1 - c + \frac{c^2}{2}$. Substituting $c = c_0 = \frac{1}{e^{\varepsilon/2} + 1}$ and $d = \frac{1}{2}$, we obtain an lower bound on the expected-error ratio, for sufficiently large N ,

$$\frac{Q^{(k)}(BRR)}{N} \geq \frac{1}{4e^{\varepsilon/2}} \left(\frac{e^{\varepsilon/2} - 1}{e^{\varepsilon/2} + 1} + 1 \right) = \frac{1}{2(e^{\varepsilon/2} + 1)}.$$

For the upper bound, substituting $c = c_0 = \frac{1}{e^{\varepsilon/2} + 1}$ and $d = \frac{c}{2}$, we have, for sufficiently large N ,

$$\frac{Q^{(k)}(BRR)}{N} \leq \frac{c^2(e^{\varepsilon} - 1) + 2(1 - c + c^2/2)}{4e^{\varepsilon/2}} = \frac{3e^{\varepsilon/2} + 2}{4(e^{\varepsilon/2} + 1)^2}.$$

Combining both bounds, we have, for sufficiently large N ,

$$\frac{1}{2(e^{\varepsilon/2} + 1)} \leq \frac{Q^{(k)}(BRR)}{N} \leq \frac{3e^{\varepsilon/2} + 2}{4(e^{\varepsilon/2} + 1)^2}.$$

(2) Next, we consider the second case: $2n < m$, which gives $0 \leq d < \frac{c}{2} < \frac{1}{4}$. We derive the asymptotic equivalent expression for the local expected error of BRR, for sufficiently large N ,

$$\begin{aligned} \frac{Q^{(k)}(BRR)}{N} &= \left[\frac{(c-d)^2}{2} + \frac{d^2}{2} \right] \cdot \frac{e^{\varepsilon}}{1 + (e^{\varepsilon} - 1)c} \\ &+ \left[\frac{(1-d)^2}{2} - \frac{(c-d)^2}{2} \right] \cdot \frac{1}{1 + (e^{\varepsilon} - 1)c} \\ &= \frac{1}{2} \cdot \frac{(1 - 2d + 2d^2)e^{\varepsilon} + (c-1)(c+1-2d)(e^{\varepsilon} - 1)}{(e^{\varepsilon} - 1)c + 1} \\ &= \frac{2e^{\varepsilon}d^2 - 2(ce^{\varepsilon} - c + 1)d + e^{\varepsilon} + (c-1)(c+1)(e^{\varepsilon} - 1)}{2((e^{\varepsilon} - 1)c + 1)}. \end{aligned}$$

For the expression $2e^{\varepsilon}d^2 - 2(ce^{\varepsilon} - c + 1)d$, substituting $c = c_0$ into the expression gives $2e^{\varepsilon}d^2 - 2e^{\frac{\varepsilon}{2}}d$. We observe that the function is decreasing over the interval $(0, \frac{c}{2})$. Now we estimate the ratio $Q^{(k)}(BRR)/N$ at the two endpoints as follows.

- When $d = 0$, corresponding to the above case (3), we have, as $N \rightarrow +\infty$, $c \rightarrow c_0$ and

$$\begin{aligned} \frac{Q^{(k)}(BRR)}{N} &\rightarrow \frac{1}{2e^{\varepsilon/2}} [e^{\varepsilon} + (c_0 + 1)(c_0 - 1)(e^{\varepsilon} - 1)] \\ &= \frac{e^{\frac{\varepsilon}{2}}}{2} - \frac{e^{\frac{\varepsilon}{2}} - 1}{2(e^{\frac{\varepsilon}{2}} + 1)} (\sqrt{e^{\varepsilon} + 2e^{\varepsilon} + 1} + 1) \\ &= \frac{1}{e^{\varepsilon/2} + 1}. \end{aligned}$$

- When $d = \frac{c}{2} \rightarrow \frac{c_0}{2}$, as in case (1), we have, as $N \rightarrow +\infty$,

$$\frac{Q^{(k)}(BRR)}{N} \rightarrow \frac{3e^{\varepsilon/2} + 2}{4(e^{\varepsilon/2} + 1)^2}.$$

Combining both bounds, we have, for sufficiently large N ,

$$\frac{3e^{\varepsilon/2} + 2}{4(e^{\varepsilon/2} + 1)^2} \leq \frac{Q^{(k)}(BRR)}{N} \leq \frac{1}{e^{\varepsilon/2} + 1}.$$

Hence, for each priori number k , the upper and lower bounds for the local expected-error ratio are obtained at the points $d = 0$ and $d = \frac{1}{2}$, respectively.

A.3 Proof of Theorem 3

Denote $n = dN$ as before. Due to the symmetry, we can account for only the case $k \leq (N+1)/2$. Since N tends to infinity, we can assume that both N and $m = m(N)$ are odd.

Denote $m_0 = \frac{N}{e^{\frac{\varepsilon}{2}} + 1}$, $c_0 = \frac{m_0}{N} = \frac{1}{e^{\frac{\varepsilon}{2}} + 1}$ and $c = c(N) = \frac{m}{N}$. Due to $\lim_{N \rightarrow \infty} m/N = \frac{1}{e^{\frac{\varepsilon}{2}} + 1}$ by Theorem 1, we have $\lim_{N \rightarrow \infty} c(N) = c_0$, that is,

$$\lim_{N \rightarrow \infty} \frac{m - m_0}{N} = \lim_{N \rightarrow \infty} \frac{m - \frac{N}{e^{\frac{\varepsilon}{2}} + 1}}{N} = \lim_{N \rightarrow \infty} \frac{m}{N} - \frac{1}{e^{\frac{\varepsilon}{2}} + 1} = 0.$$

We are now ready to calculate the global expected error of BRR with the dynamic splitting number $m = cN$ as follows.

$$\begin{aligned} N \cdot Q_g(BRR) &= \sum_{k=1}^N Q^{(k)}(BRR) \\ &= 2 \left(\sum_{k=1}^{\frac{m+1}{2}} Q^{(k)}(BRR) + \sum_{k=\frac{m+3}{2}}^{\frac{N+1}{2}} Q^{(k)}(BRR) \right) - Q^{(\frac{N+1}{2})}(BRR) \\ &= \sum_{n=0}^{\frac{m-1}{2}} \frac{2e^{\varepsilon}}{N} \frac{n(n+1) - 2n + c(e^{\varepsilon} - 1)(Nc - 1 - 2n) + N - 1}{1 + (e^{\varepsilon} - 1)c} \\ &\quad + \sum_{n=\frac{m+1}{2}}^{\frac{N-1}{2}} \frac{N \left[\left(c^2 - \frac{1}{N^2} \right) (e^{\varepsilon} - 1) + 2 - \frac{4n}{N} + \frac{4n}{N^2} + \frac{4n^2}{N^2} - \frac{2}{N} \right]}{2(1 + (e^{\varepsilon} - 1)c)} \\ &\quad - \frac{N \cdot \left(\left(c^2 - \frac{1}{N^2} \right) (e^{\varepsilon} - 1) + 2 - \frac{1}{N^2} - 1 + \frac{2}{N} - \frac{2}{N} \right)}{4(1 + (e^{\varepsilon} - 1)c)}. \end{aligned} \tag{22}$$

Denote $A = \frac{1}{(e^{\varepsilon} - 1)c + 1}$, then

$$\begin{aligned}
& N \cdot Q_g(BRR) \\
&= A \sum_{n=0}^{\frac{cN-1}{2}} [c(e^\epsilon - 1)(Nc - 1) + N - 1] \\
&\quad + A \sum_{n=0}^{\frac{cN-1}{2}} \left[\left(\frac{2}{N} - 2 \right) e^\epsilon - 2(c-1)(e^\epsilon - 1) \right] n + A \sum_{n=0}^{\frac{cN-1}{2}} \frac{2e^\epsilon}{N} n^2 \\
&\quad + \frac{A}{2} \sum_{n=\frac{cN+1}{2}}^{\frac{N-1}{2}} N \left(\left(c^2 - \frac{1}{N^2} \right) (e^\epsilon - 1) + 2 - \frac{2}{N} \right) \\
&\quad + \frac{A}{2} \sum_{n=\frac{cN+1}{2}}^{\frac{N-1}{2}} \left[(-4)n + \frac{4n}{N} \right] + \frac{A}{2} \sum_{n=\frac{cN+1}{2}}^{\frac{N-1}{2}} \frac{4}{N} n^2 \\
&\quad - \frac{A}{4} N \cdot \left(\left(c^2 - \frac{1}{N^2} \right) (e^\epsilon - 1) + 1 - \frac{1}{N^2} \right).
\end{aligned}$$

By simplifying the above summation, the result is

$$\begin{aligned}
f(c) &= \frac{(e^\epsilon - 1)c^3 + 3(e^\epsilon - 1)c^2 - 3e^\epsilon + 4}{12[(e^\epsilon - 1)c + 1]} N^2 - \frac{1}{12} \\
&= \frac{(e^\epsilon - 1)N^2c^3 + 3(e^\epsilon - 1)N^2c^2 - (e^\epsilon - 1)c - 3e^\epsilon - 1 + 4N^2}{12[(e^\epsilon - 1)c + 1]}.
\end{aligned} \tag{23}$$

This approaches $\frac{7e^{\frac{\epsilon}{2}} + 9}{12(e^{\frac{\epsilon}{2}} + 1)^2}$ when N tends to infinity, by using $\lim_{N \rightarrow \infty} c(N) = c_0 = \frac{1}{e^{\epsilon/2} + 1}$.

Thus, we come to the final conclusion,

$$\lim_{N \rightarrow \infty} \frac{Q_g(BRR, N)}{N} = \lim_{N \rightarrow \infty} \frac{N^2(7e^{\epsilon/2} + 9)}{12(e^{\epsilon/2} + 1)^2} \bigg/ N^2 = \frac{7e^{\epsilon/2} + 9}{12(e^{\epsilon/2} + 1)^2}.$$

We would like to mention that, the limit of ratios depends directly on the terms of order N^2 in the expansion of (22), then the case of even N or m remains the final result while the slight changes on the upper and lower limits of the sums take place, like replacing $\frac{m-1}{2}$ by $\frac{m}{2} - 1$.

Besides, for a priori position k , $Q^{(k)}(GRR, N) \sim (1 - 2d + 2d^2)N/2$. Then,

$$\begin{aligned}
N \cdot Q_g(GRR) &= \sum_{k=1}^N Q^{(k)}(GRR) \sim 2 \sum_{n=1}^{\frac{N-1}{2}} \left(\frac{N}{2} - n + \frac{n^2}{N} \right) \\
&= 2 \left(\frac{N^2 - N}{4} - \frac{N^2 - 1}{8} + \frac{N^2 - 1}{24} \right) \\
&= \frac{1}{3} N^2 - \frac{1}{2} N + \frac{1}{6}.
\end{aligned}$$

This gives that $\lim_{N \rightarrow \infty} Q_g(GRR, N)/N = 1/3$, which is always no less than the limiting ratio for BRR, since

$$\frac{7e^{\epsilon/2} + 9}{12(e^{\epsilon/2} + 1)^2} \leq \frac{8e^{\epsilon/2} + 8}{12(e^{\epsilon/2} + 1)^2} = \frac{2}{3(e^{\epsilon/2} + 1)} \leq \frac{1}{3}.$$

B Additional Experimental Evaluation

B.1 BRR with Jaccard-Similarity Utility

Following the setting of Section 3, we conduct the experiments in the equidistant point domain $\mathcal{X} = \{1, 2, \dots, N\}$ with varying N . We denote data utility λ by the pairwise generalized Jaccard similarity coefficient (Tinimoto coefficient), for any candidate y , given the true point x ,

$$\lambda(x, y) = \frac{xy}{x^2 + y^2 - xy}. \tag{24}$$

The global utility Q_g is defined by (14) for comparisons among BRR, GRR, Geometric and Exponential mechanisms. In this scenario, the larger Q_g gives the better performance. We compare and analyze the performance of three different differential privacy protection mechanisms: BRR, GRR, Geometric and the Exponential mechanism, with privacy budgets and candidates number N .

From Fig. 11, we observe that, given N , the utility increases with growing privacy budgets across four mechanisms. This trend is inherent to the properties of differential privacy. When examined vertically, it is evident that the global expected error of the BRR remains relatively stable across different N for given ϵ and is much lower than those of the other three mechanisms. These findings indicate that the BRR mechanism enhances data utility largely. This substantiates our assertions, cf. Theorem 4.

B.2 BRR in Private GBDT

Now, we turn to the applications of BRR in continuous domains for machine learning. The current experiments are conducted by deploying BRR mechanism as an alternative to the Laplace mechanism for perturbing each leaf value in DPBoost by Li et al. [18]. Originally, for training DPBoost, the sensitivity of the noise added after each geometric leaf clipping is $\Delta q = \min(\frac{g_l^*}{1+\lambda}, 2g_l^*(1-\eta)^{t-1})$. In the training of new DP-GBDT called DPBoost-BRR due to the above alternative, the perturbation range of leaf clipping should be $[-c, c]$ with $c = \Delta q/2$. Then the range $[-c, c]$ is equally divided for suitable L that gives $n = \frac{2c}{L} + 1$ discrete equidistant points. The leaf node values are first adjusted to the closest candidate. Finally, applying the BRR mechanism gives the perturbed leaf values.

For the regression tasks on the Bias, abalone and superconduct (sup) datasets, RMSE (Root Mean Squared Error) is used as the evaluation metric. The number of instances and features for these datasets are (4177, 8), (7588, 23), and (21263, 82), respectively. We compare the performance of applying BRR and Laplace mechanisms, respectively, in DPBoost under different privacy budgets. As shown in Fig. 12, DPBoost-BRR consistently outperforms DPBoost in terms of regression performance. Moreover, the BRR mechanism

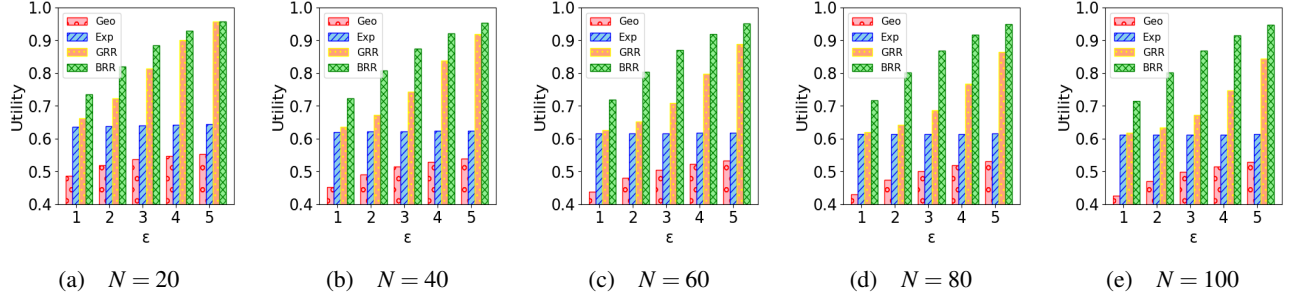


Figure 11: Comparisons of BRR to GRR, Geometric mechanism (Geo) and Exponential mechanism for Jaccard-similarity defined utility with varying privacy budgets and count of candidates.

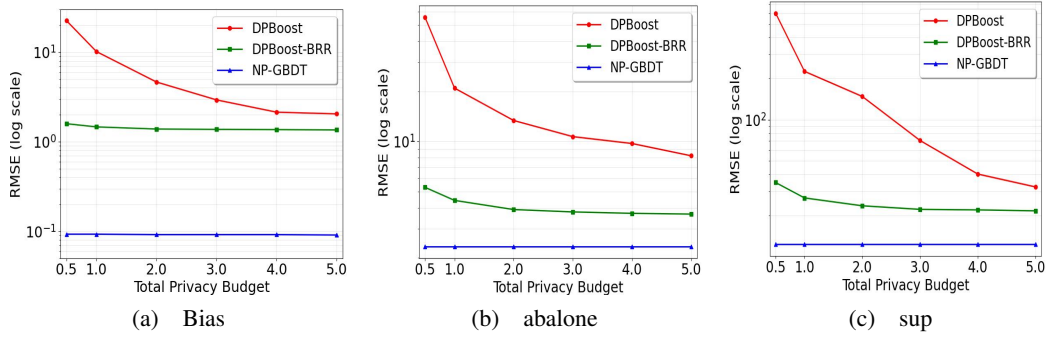


Figure 12: RMSE comparison of BRR and Laplace mechanisms in privacy-preserving decision tree models with varying total privacy budgets, cf. DPBoost [18] (using Laplace), DPBoost-BRR (using BRR), NP-GBDT (the vanilla GBDT) [18].

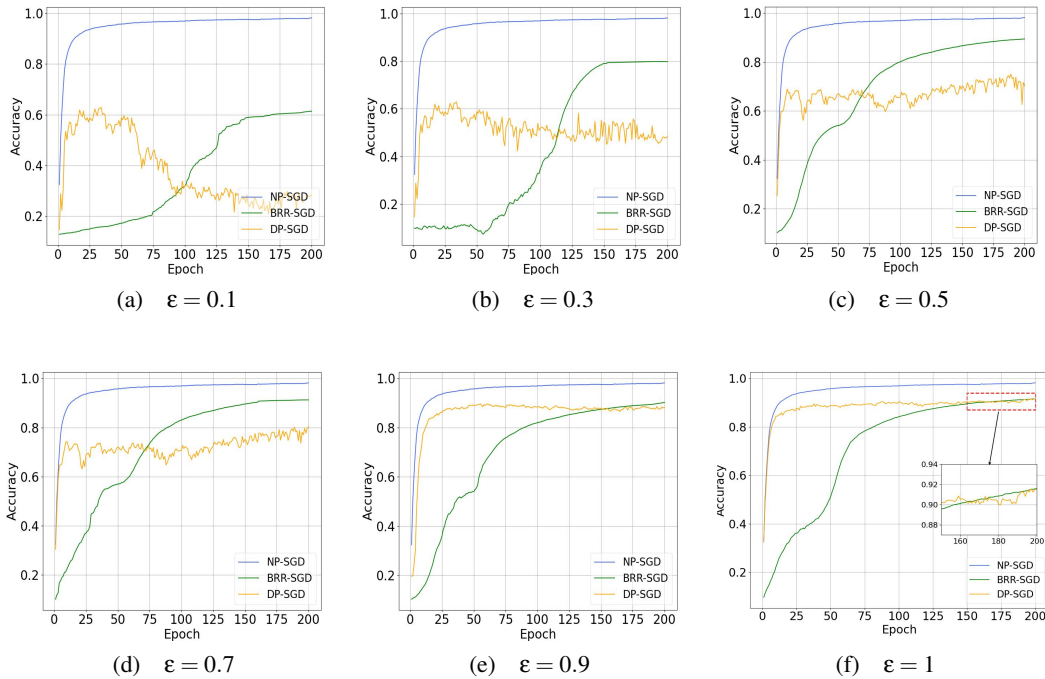


Figure 13: Accuracy dynamics of differentially private SGD models with Laplace [1] and BRR mechanisms, respectively, compared to Non-Private SGD (NP-SGD) across training iterations under varying total privacy budgets, cf. DP-SGD [1] (using Laplace), BRR-SGD (using BRR).

demonstrates greater stability across various privacy budgets, highlighting its advantage in balancing privacy preservation and model accuracy. For small privacy budgets, the RMSE of DPBoost is excessively high, weakening its practicality in prediction. Even at $\epsilon = 5$, the RMSE of DPBoost-BRR achieves about 66%, 45%, and 67% of that for DPBoost (with Laplace) on three datasets, respectively.

B.3 BRR in Private SGD

The experiment is conducted based on the differential private SGD (DP-SGD) [1] involving Laplace mechanism. The BRR mechanism is compared with the Laplace mechanism by their alternative in DP-SGD, with mainly analyzing how these two obfuscation mechanisms affect model accuracy under varying privacy budgets. Following the procedure in [1], gradients at each iteration are clipped with the scalar norm constriction at most C , thereby bounding the sensitivity of the algorithm by C . To better adapt to the changing gradient scale during training, the clipping threshold C is made adaptive. Specifically, let $\mathbf{g}_t$ be the mean gradient norm of the current mini-batch at iteration t , then C is updated as $C_t = \max(\rho \cdot C_{t-1} + (1 - \rho) \cdot \mathbf{g}_t, C_{\min})$, where $\rho \in (0, 1)$ is the decay rate for exponential moving averaging, and $C_{\min}$ is a predefined minimum threshold to prevent C_t from being too small and causing numerical instability. NP-SGD is the base scheme where no noise is added on SGD. For the BRR-SGD, after the gradient clipping, the interval $[-C_t, 0]$ or $[0, C_t]$ is split accordingly for the discrete probability distribution on equidistant points of BRR as before.

The experiments are carried out on the MNIST dataset, a widely-used benchmark for handwritten digit recognition, for the classification task. The model performance is evaluated using accuracy as the primary metric, the prediction results of the model are compared with the real labels of the test set and the accuracy is output, with the aim of assessing the effectiveness of these noise mechanisms across different privacy budget conditions.

As shown in Fig. 13, with smaller privacy budgets, the injected noise increases, resulting in a decline in the accuracy of model updates. Under these conditions, the Laplace mechanism introduces larger noise magnitudes, leading to significant fluctuations in accuracy and difficulty in maintaining stability. In contrast, the BRR mechanism demonstrates better stability, although its convergence is slower. As the privacy budget increases, the noise intensity decreases, reducing its impact on model updates. The Laplace mechanism can achieve its potential accuracy more quickly; however, its fluctuations persist. On the other hand, the BRR mechanism exhibits a more stable convergence process and eventually reaches, or even surpasses, the accuracy of the Laplace mechanism. When the privacy budget is low, BRR-SGD is significantly better than DP-SGD, and the accuracy indicators are improved by about 28%, 21%, and 12%, for the privacy

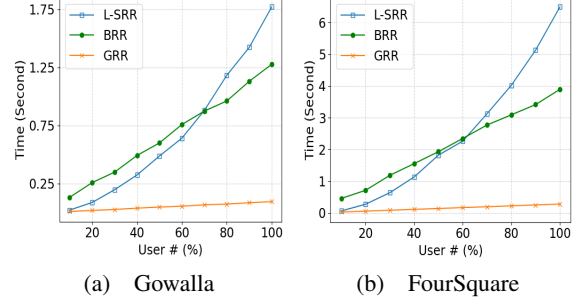


Figure 14: Runtime of L-SRR [29], GRR [15], and BRR for the server with varying the number of users.

budgets 0.3, 0.5, and 0.7, respectively. At higher privacy budgets, both mechanisms inject smaller amounts of noise, and the model’s performance approximates a noise-free scenario. At this phase, the gap in accuracy between DP-SGD and BRR-SGD narrows, and both ultimately achieve similarly high accuracy levels.

Overall, the BRR mechanism proves to be more advantageous in terms of stability and final accuracy of SGD, particularly on balancing privacy protection with maintaining strong model performance.

B.4 Runtime

In this section for runtime, the testing environment consists of the Intel64 Family 6 Model 158 CPU running at 3.41 GHz, 16 GB main memory, Windows 10 version 21H2, and Python version 3.12.

Since each user perturb their locations locally, the user-side runtime is only 0.04 second for each user on average. Then we mainly report the server-side runtime by testing 10% to 100% of each dataset demonstrated in Fig. 8 with comparisons to GRR and L-SRR. For example, “10%” means that the number of users whose locations are to be perturbed accounts to 10% of locations in the whole domain for each dataset. The division of the whole domain can be finished ahead and is not involved in this testing runtime.

Fig. 14 shows that GRR takes a little time due to the simple process while L-SRR consumes more time than BRR due to its complex computation for multiple groups. The runtime of BRR grows more slowly than L-SRR as the number of users reaches 3000 (e.g., 4 seconds for FourSquare), which is acceptable in real-life applications.